The Cross-Cultural Big Two: Assessment of a Culturally De-Centered Personality Inventory

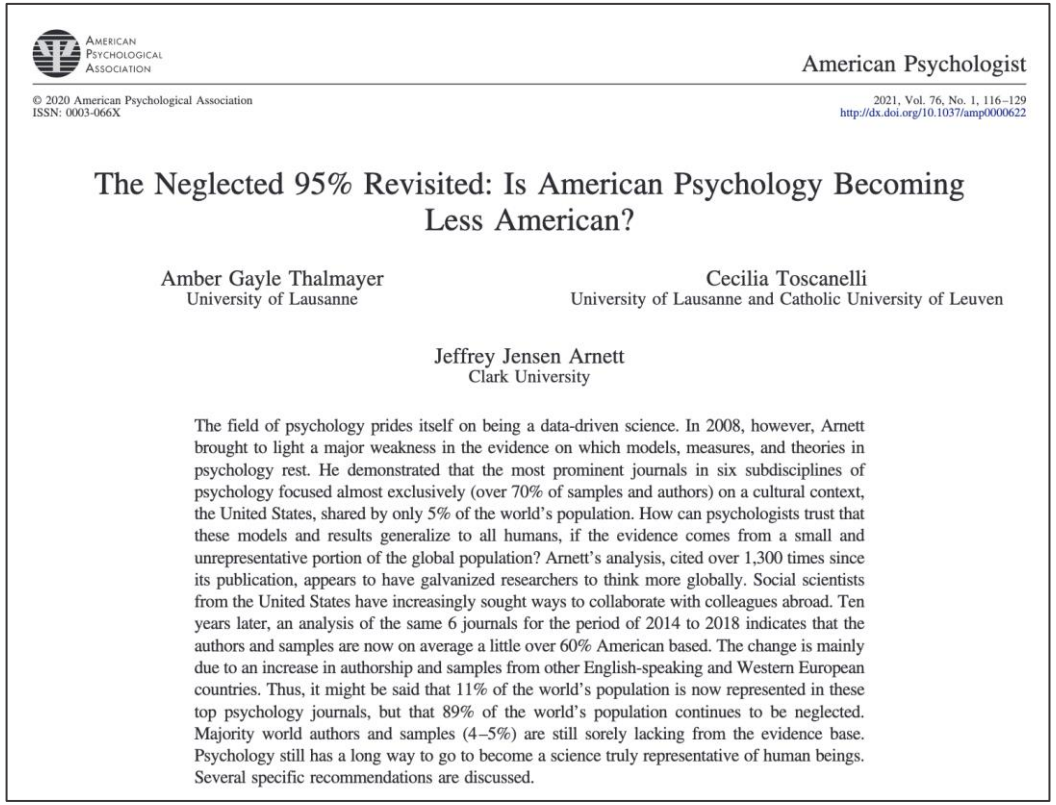
Kendall Mather[1], David Condon[1], Gerard Saucier[1], & Amber Gayle Thalmayer[2]

[1]University of Oregon

[2]University of Zurich

Cultural representation in psychology research

A recent assessment of representation in psychology research by Thalmayer et al. (2020) ….



American Psychological Association

American Psychologist

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ISSN: 0003-066X

2021, Vol. 76, No. 1, 116–129
http://dx.doi.org/10.1037/amp0000622

The Neglected 95% Revisited: Is American Psychology Becoming Less American?

Amber Gayle Thalmayer
University of Lausanne

Cecilia Toscanelli
University of Lausanne and Catholic University of Leuven

Jeffrey Jensen Arnett
Clark University

The field of psychology prides itself on being a data-driven science. In 2008, however, Arnett brought to light a major weakness in the evidence on which models, measures, and theories in psychology rest. He demonstrated that the most prominent journals in six subdisciplines of psychology focused almost exclusively (over 70% of samples and authors) on a cultural context, the United States, shared by only 5% of the world's population. How can psychologists trust that these models and results generalize to all humans, if the evidence comes from a small and unrepresentative portion of the global population? Arnett's analysis, cited over 1,300 times since its publication, appears to have galvanized researchers to think more globally. Social scientists from the United States have increasingly sought ways to collaborate with colleagues abroad. Ten years later, an analysis of the same 6 journals for the period of 2014 to 2018 indicates that the authors and samples are now on average a little over 60% American based. The change is mainly due to an increase in authorship and samples from other English-speaking and Western European countries. Thus, it might be said that 11% of the world's population is now represented in these top psychology journals, but that 89% of the world's population continues to be neglected. Majority world authors and samples (4–5%) are still sorely lacking from the evidence base. Psychology still has a long way to go to become a science truly representative of human beings. Several specific recommendations are discussed.

- Some improvement in representation since 2008

"Majority world authors and samples are still sorely lacking from the evidence base"

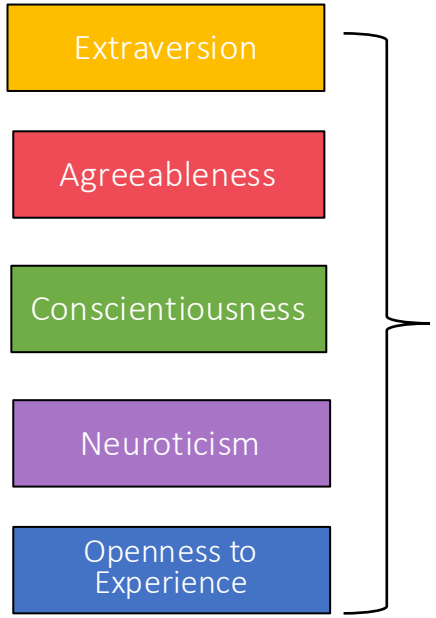
About 89% of population is still neglected in psychology research

- Mostly from societies in Asia, Africa, Latin America, and the Caribbean

Cultural representation in “universal” personality assessment models

Cultural representation in “universal” personality assessment models

Big Five/Five Factor Model



- Suggested to be universal (Costa & McCrae, 1992) but derived using western samples
- Structure not spontaneously found in many lexical studies

Cultural representation in “universal” personality assessment models

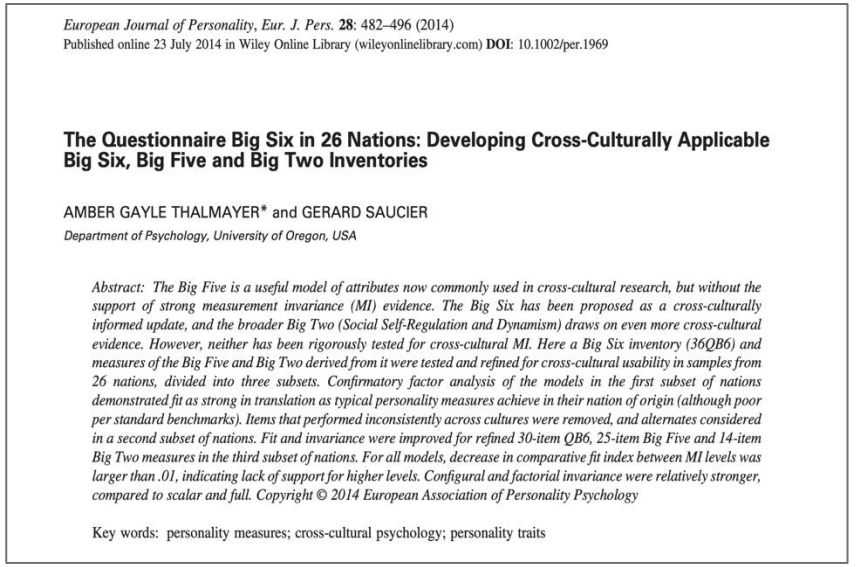
Big Five/Five Factor Model

- Extraversion
- Agreeableness
- Conscientiousness
- Neuroticism
- Openness to Experience

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Big Six

- Honesty/humility
- Emotional Stability
- Extraversion
- Agreeableness
- Conscientiousness
- Openness to Experience



European Journal of Personality, Eur. J. Pers. **28**: 482–496 (2014)
Published online 23 July 2014 in Wiley Online Library (wileyonlinelibrary.com) **DOI**: 10.1002/per.1969

The Questionnaire Big Six in 26 Nations: Developing Cross-Culturally Applicable Big Six, Big Five and Big Two Inventories

AMBER GAYLE THALMAYER* and GERARD SAUCIER

Department of Psychology, University of Oregon, USA

Abstract: The Big Five is a useful model of attributes now commonly used in cross-cultural research, but without the support of strong measurement invariance (MI) evidence. The Big Six has been proposed as a cross-culturally informed update, and the broader Big Two (Social Self-Regulation and Dynamism) draws on even more cross-cultural evidence. However, neither has been rigorously tested for cross-cultural MI. Here a Big Six inventory (36QB6) and measures of the Big Five and Big Two derived from it were tested and refined for cross-cultural usability in samples from 26 nations, divided into three subsets. Confirmatory factor analysis of the models in the first subset of nations demonstrated fit as strong in translation as typical personality measures achieve in their nation of origin (although poor per standard benchmarks). Items that performed inconsistently across cultures were removed, and alternates considered in a second subset of nations. Fit and invariance were improved for refined 30-item QB6, 25-item Big Five and 14-item Big Two measures in the third subset of nations. For all models, decrease in comparative fit index between MI levels was larger than .01, indicating lack of support for higher levels. Configural and factorial invariance were relatively stronger, compared to scalar and full. Copyright © 2014 European Association of Personality Psychology

Key words: personality measures; cross-cultural psychology; personality traits

- More support for universality of the big six structure found in lexical studies
- Some dimensions still lack relevance in many contexts

Cultural representation in “universal” personality assessment models

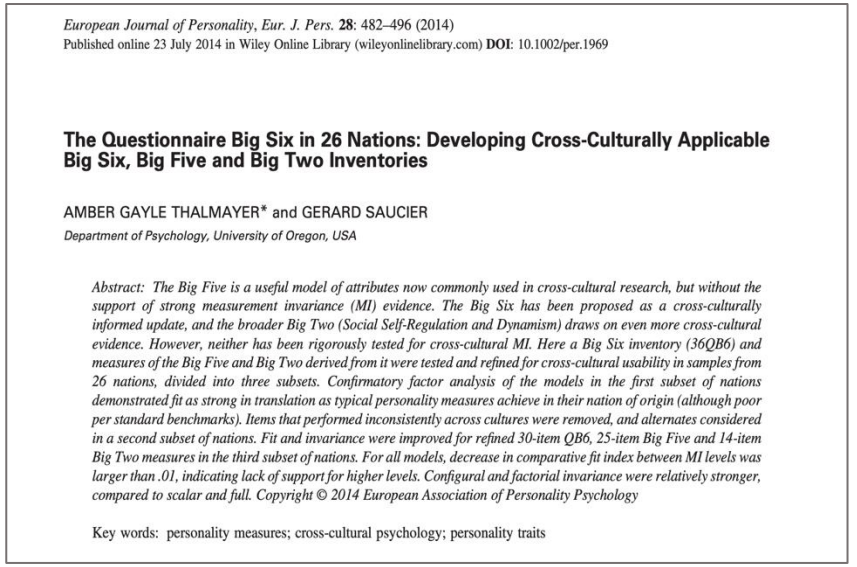
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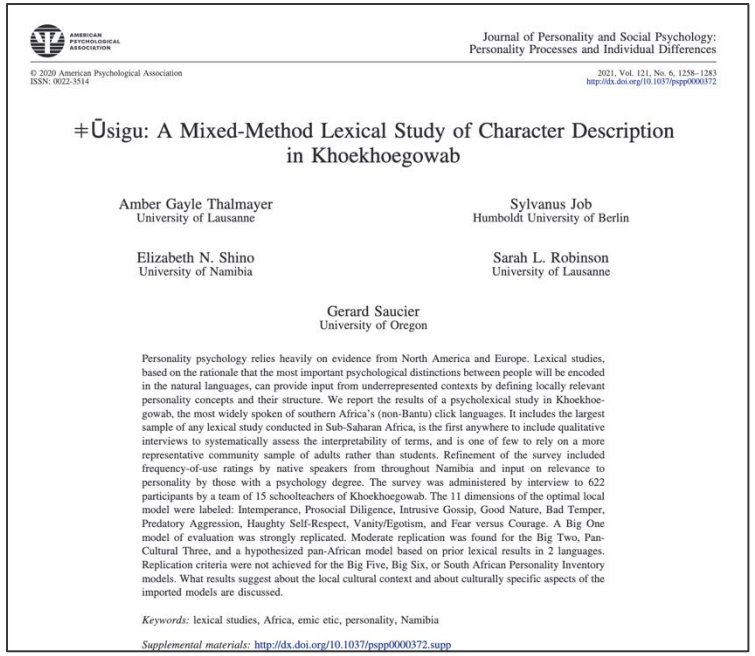
- More support for universality of the big six structure found in lexical studies
- Some dimensions still lack relevance in many contexts

A lack of scalar-level measurement invariance evidence for popular measures prohibits comparison of trait-scores across cultures (Thalmayer & Saucier, 2014)

Local personality structures found in lexical studies

- Structure of personality defined based on the most locally meaningful personality concepts
- Number and nature of traits vary widely across contexts

Culture specific (emic) models:



Journal of Personality and Social Psychology: Personality Processes and Individual Differences

© 2020 American Psychological Association
ISSN: 0022-3514

2021, Vol. 121, No. 6, 1258–1283
http://dx.doi.org/10.1037/pspp0000372

ǂŪsigu: A Mixed-Method Lexical Study of Character Description in Khoekhoegowab

Amber Gayle Thalmayer
University of Lausanne

Sylvanus Job
Humboldt University of Berlin

Elizabeth N. Shino
University of Namibia

Sarah L. Robinson
University of Lausanne

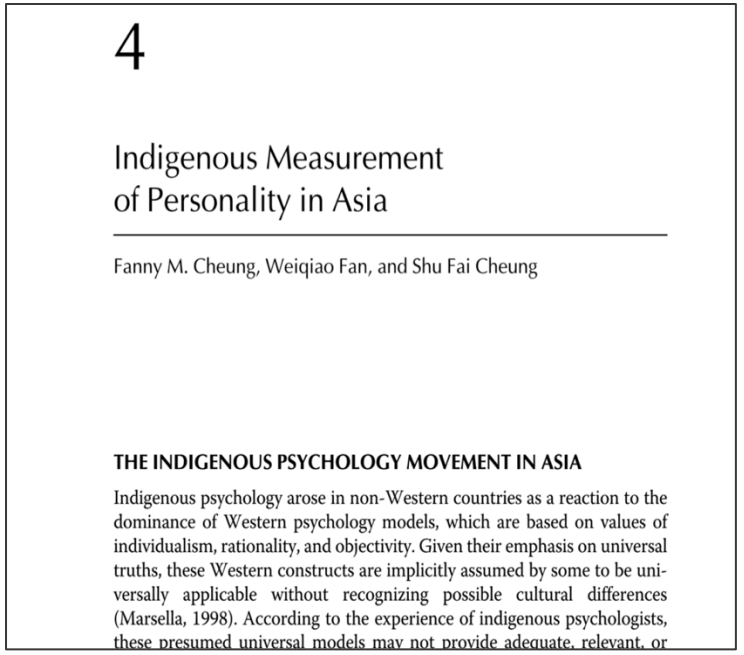
Gerard Saucier
University of Oregon

Personality psychology relies heavily on evidence from North America and Europe. Lexical studies, based on the rationale that the most important psychological distinctions between people will be encoded in the natural languages, can provide input from underrepresented contexts by defining locally relevant personality concepts and their structure. We report the results of a psycholexical study in Khoekhoegowab, the most widely spoken of southern Africa's (non-Bantu) click languages. It includes the largest sample of any lexical study conducted in Sub-Saharan Africa, is the first anywhere to include qualitative interviews to systematically assess the interpretability of terms, and is one of few to rely on a more representative community sample of adults rather than students. Refinement of the survey included frequency-of-use ratings by native speakers from throughout Namibia and input on relevance to personality by those with a psychology degree. The survey was administered by interview to 622 participants by a team of 15 schoolteachers of Khoekhoegowab. The 11 dimensions of the optimal local model were labeled: Intemperance, Prosocial Diligence, Intrusive Gossip, Good Nature, Bad Temper, Predatory Aggression, Haughty Self-Respect, Vanity/Egotism, and Fear versus Courage. A Big One model of evaluation was strongly replicated. Moderate replication was found for the Big Two, Pan-Cultural Three, and a hypothesized pan-African model based on prior lexical results in 2 languages. Replication criteria were not achieved for the Big Five, Big Six, or South African Personality Inventory models. What results suggest about the local cultural context and about culturally specific aspects of the imported models are discussed.

Keywords: lexical studies, Africa, emic etic, personality, Namibia

Supplemental materials: http://dx.doi.org/10.1037/pspp0000372.supp

Thalmayer et al. (2021)



4

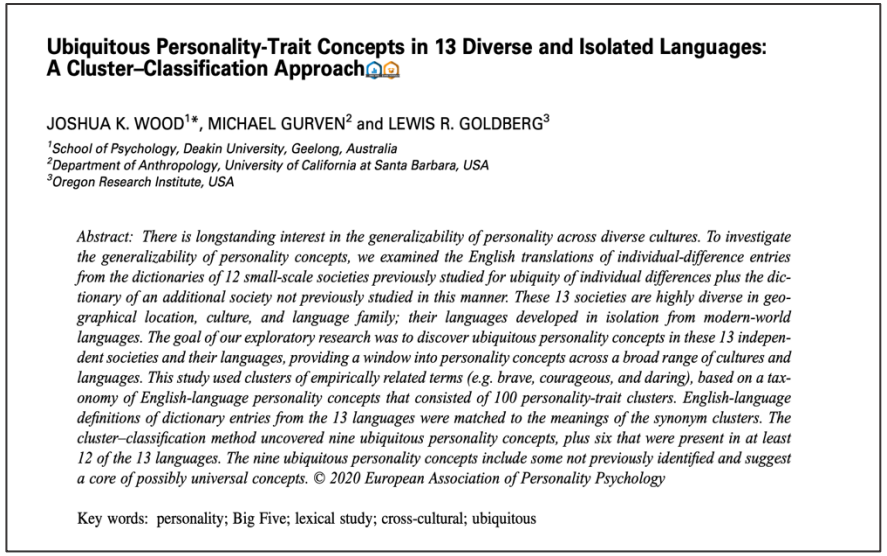
Indigenous Measurement of Personality in Asia

Fanny M. Cheung, Weiqiao Fan, and Shu Fai Cheung

THE INDIGENOUS PSYCHOLOGY MOVEMENT IN ASIA

Indigenous psychology arose in non-Western countries as a reaction to the dominance of Western psychology models, which are based on values of individualism, rationality, and objectivity. Given their emphasis on universal truths, these Western constructs are implicitly assumed by some to be universally applicable without recognizing possible cultural differences (Marsella, 1998). According to the experience of indigenous psychologists, these presumed universal models may not provide adequate, relevant, or

Cheung et al. (2017)



Ubiquitous Personality-Trait Concepts in 13 Diverse and Isolated Languages: A Cluster–Classification Approach

JOSHUA K. WOOD[1]*, MICHAEL GURVEN[2] and LEWIS R. GOLDBERG[3]

[1]*School of Psychology, Deakin University, Geelong, Australia*
[2]*Department of Anthropology, University of California at Santa Barbara, USA*
[3]*Oregon Research Institute, USA*

Abstract: There is longstanding interest in the generalizability of personality across diverse cultures. To investigate the generalizability of personality concepts, we examined the English translations of individual-difference entries from the dictionaries of 12 small-scale societies previously studied for ubiquity of individual differences plus the dictionary of an additional society not previously studied in this manner. These 13 societies are highly diverse in geographical location, culture, and language family; their languages developed in isolation from modern-world languages. The goal of our exploratory research was to discover ubiquitous personality concepts in these 13 independent societies and their languages, providing a window into personality concepts across a broad range of cultures and languages. This study used clusters of empirically related terms (e.g. brave, courageous, and daring), based on a taxonomy of English-language personality concepts that consisted of 100 personality-trait clusters. English-language definitions of dictionary entries from the 13 languages were matched to the meanings of the synonym clusters. The cluster–classification method uncovered nine ubiquitous personality concepts, plus six that were present in at least 12 of the 13 languages. The nine ubiquitous personality concepts include some not previously identified and suggest a core of possibly universal concepts. © 2020 European Association of Personality Psychology

Key words: personality; Big Five; lexical study; cross-cultural; ubiquitous

Wood et al. (2020)

Local personality structures found in lexical studies

ǂŪsigu: A mixed-method lexical study of character description in Khoekhoegowab



Thalmayer, A. G., Job, S., Shino, E. N., Robinson, S. L., & Saucier, G. (2021). ǂŪsigu: A mixed-method lexical study of character description in Khoekhoegowab. *Journal of Personality and Social Psychology*, *121*(6), 1258.

- A few of the 26 dimensions….

au (n = 10)	anxious/uneasy, feel	(1) bitter, negative, resentful, stingy, antisocial; (2) tough, austere, hardy
oaxaedīsen (n = 10)	unattached/eligible for marriage, act/behave as if	to dress up, make oneself beautiful or look good in order to stand out or attract the attention of others – especially with regard to the opposite sex
dî!nâxa (n = 10)	cross-examine, inquire; examine (oral), tend to, inquisitive	someone who is inquisitive, who asks lots of questions in a persistent manner (curious [+]; nosy, prying [-])
ǂūxa (n = 11)	eccentric, freakish; bad=-mannered; moody; antisocial	(1) bad mannered, negative attitude that repulses others, lack of respect; (2) bad-ass, excellent, deserves credit for something; (3) good mannered, respectful. (Depends on context)
supudana (n = 10)	bright, quick to grasp, quick in/on the uptake (esp of: child)	(1) gullible, easily manipulated, too easily convinced, persuaded, or influenced (naïve); (2) intelligent, quick to grasp, learns and understands easily.
ǁgoaraxa (n = 10)	blackmail, browbeat, (prone to)	someone who wants to be begged by others, creates situations where others are in a position to beg, withholds in order to be begged.
ǂkhāxa (n = 8)	non-compliant; likely to refuse/etc.	resistant, refusal to do, help, give, etc, stubborn, uncooperative
ǂâi!gâxa (n = 9)	inclined to rethink; thoughtful; heedful, meditative, pensive; strong-minded.	(1) someone who reflects and considers before acting; (2) someone who reflects on a past action with regret or remorse, then corrects or apologizes.
!gamǂâixa (n = 7)	deep-thinking, contemplative, meditative	a deep, complex thinker, contemplative, a prophet, visionary (associated with being

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Comparison of personality across cultures

“Universal”, culturally agnostic models ☑

- Prioritize parsimony, generalizability
- Less accurate in any specific context

Culture specific models

- Prioritize accuracy in one context
- Less generalizable to outside contexts

Universal models align more closely with the goals of research involving cross-cultural comparison of personality on the same attributes and measures.

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Universal models align more closely with the goals of research involving cross-cultural comparison of personality on the same attributes and measures.

We aim to develop a culturally de-centered starting point for cross-cultural personality assessment

A candidate model: the big two

Social Self-regulation

Propriety, Socialization, Community

Dynamism

Activity, Potency, Ascendency

Dimensions map onto constructs from other theories…

Stability	Agreeableness	Communion	Getting along
Plasticity	Extraversion	Agency	Getting ahead
DeYoung (2006)	**FFM/Big five**	**Bakan (1961)** **Wiggins (1991)**	**Hogan (1983)**

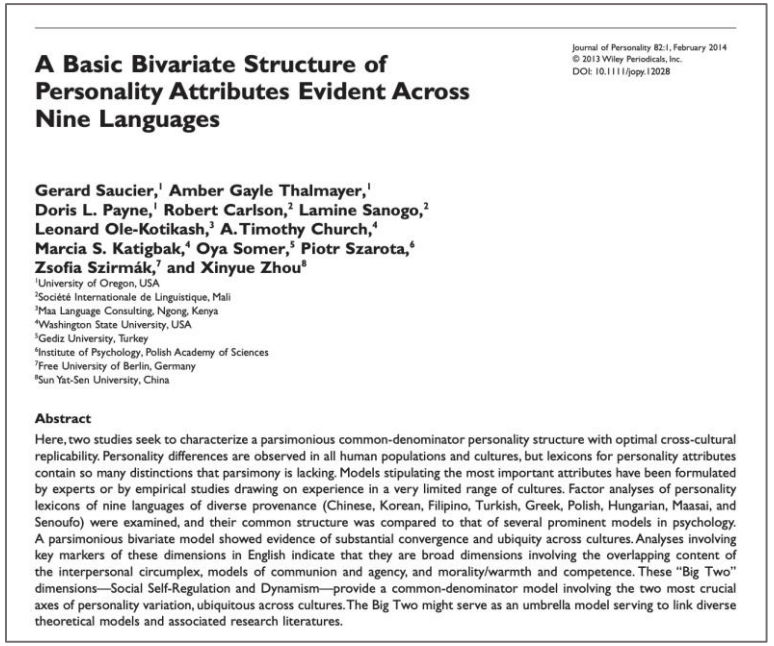
A candidate model: the big two

Social Self-regulation

Propriety, Socialization, Community

Dynamism

Activity, Potency, Ascendency



Journal of Personality 82:1, February 2014
© 2013 Wiley Periodicals, Inc.
DOI: 10.1111/jopy.12028

A Basic Bivariate Structure of Personality Attributes Evident Across Nine Languages

Gerard Saucier,[1] Amber Gayle Thalmayer,[1] Doris L. Payne,[1] Robert Carlson,[2] Lamine Sanogo,[2] Leonard Ole-Kotikash,[3] A. Timothy Church,[4] Marcia S. Katigbak,[4] Oya Somer,[5] Piotr Szarota,[6] Zsofia Szirmák,[7] and Xinyue Zhou[8]

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[2]Société Internationale de Linguistique, Mali
[3]Maa Language Consulting, Ngong, Kenya
[4]Washington State University, USA
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[6]Institute of Psychology, Polish Academy of Sciences
[7]Free University of Berlin, Germany
[8]Sun Yat-Sen University, China

Abstract

Here, two studies seek to characterize a parsimonious common-denominator personality structure with optimal cross-cultural replicability. Personality differences are observed in all human populations and cultures, but lexicons for personality attributes contain so many distinctions that parsimony is lacking. Models stipulating the most important attributes have been formulated by experts or by empirical studies drawing on experience in a very limited range of cultures. Factor analyses of personality lexicons of nine languages of diverse provenance (Chinese, Korean, Filipino, Turkish, Greek, Polish, Hungarian, Maasai, and Senoufo) were examined, and their common structure was compared to that of several prominent models in psychology. A parsimonious bivariate model showed evidence of substantial convergence and ubiquity across cultures. Analyses involving key markers of these dimensions in English indicate that they are broad dimensions involving the overlapping content of the interpersonal circumplex, models of communion and agency, and morality/warmth and competence. These "Big Two" dimensions—Social Self-Regulation and Dynamism—provide a common-denominator model involving the two most crucial axes of personality variation, ubiquitous across cultures. The Big Two might serve as an umbrella model serving to link diverse theoretical models and associated research literatures.

Dimensions map onto constructs from other theories…

Stability	Agreeableness	Communion	Getting along
Plasticity	Extraversion	Agency	Getting ahead
DeYoung (2006)	**FFM/Big five**	**Bakan (1961)** **Wiggins (1991)**	**Hogan (1983)**

Support for the big two

- First two factors derived in lexical studies across several diverse contexts (Saucier et al., 2014)
- Dimensions emerge across diverse variable selection procedures (Saucier, 1997)

A candidate model: the big two

Social Self-regulation

Propriety, Socialization, Community

Dynamism

Activity, Potency, Ascendency

Language Group	Language	Variable Selection	Rater
Korean	Korean	Restricted	Self
Finno-Ugric	Hungarian	Restricted	Self
Indo-European	Polish	Restricted	Peer
	Greek	Broad	Self
Altaic	Turkish	Broad	Self
Sino-Tibetan	Chinese	Broad	Peer
Austronesian	Filipino	Broad	Self
	Malay	Broad	Self
Nilotic	Maasai	Very Broad	Peer
Niger-Congo	Supyire-Senufo	Very Broad	Peer
Khoi-san	KKG	Broad	Peer

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Support for the big two

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- Dimensions emerge across diverse variable selection procedures (Saucier, 1997)

The present work: develop a measure from an initial pool of 55 items

- These 55 items were identified based on several past (Saucier et al., 2014) and forthcoming lexical studies.

30 Social Self-Regulation items:			
Tend to be…..	"disobedient" (-),	"hot-tempered" (-),	"obedient",
"aggressive" (-)	"faithful",	"hotheaded" (-),	"peaceful",
"arrogant" (-)	"friendly",	"humane",	"polite",
"bad" (-)	"generous"	"industrious",	"respectful"
"benevolent",	"gentle",	"inhumane" (-),	
"calm",	"good-natured",	"kind",	
"compassionate",	"gossipy" (-),	"kindhearted",	
"courteous"	"hardworking",	"modest",	
"diligent"	"honest",	"naïve",	

25 Dynamism items:		
Tend to be…..	"courageous",	"quiet" (-),
"a loner" (-),	"cowardly" (-),	"sad" (-),
"active",	"daring",	"self-confident",
"bashful" (-),	"determined",	"shy" (-),
"bold",	"dynamic",	"silent" (-),
"brave",	"energetic",	"timid" (-),
"cheerful",	"happy",	"unhappy" (-),
"clever",	"lively",	"weak" (-)
"confident",	"Pessimistic" (-)	

The present work: develop a measure from an initial pool of 55 items

☑ Step 1 • Develop preliminary item pool

The present work: develop a measure from an initial pool of 55 items

☑ Step 1
- Develop preliminary item pool

☑ Step 2

<u>Quantitative analyses (present work)</u>

- Item and scale-level analyses candidate items
- Select items to drop from the final scales

Qualitative work (in progress)

- Identify items that are unfamiliar in sample of African nations

The present work: develop a measure from an initial pool of 55 items

☑ **Step 1**
- Develop preliminary item pool

☑ **Step 2**

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- Item and scale-level analyses candidate items
- Select items to drop from the final scales

Qualitative work (in progress)
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☑ **Step 3**
- Validation of scales against external criteria and other two factor models

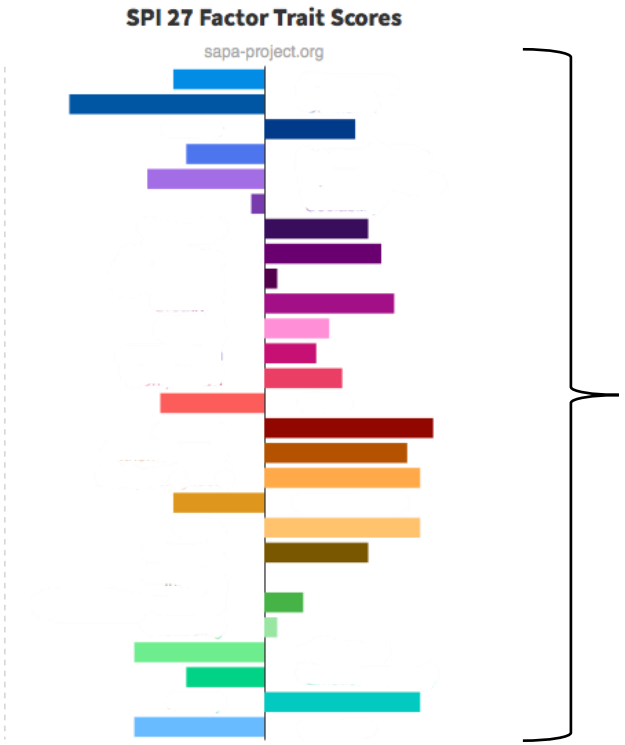
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☑ **Step 4**
- Validation of scales using Africa Long Life Study data (African community sample)

Dataset and participants

The SAPA Project

- Online data collection tool that uses a planned missingness design (~90% missingness)
- Participants are administered a random subset of items



Participants receive information about their personalities in exchange for participation

Participants:

Initial sample: N = 1,377,047

Exclusions:

- Under 14 or over 90 years of age N = 277,879
- Did not report country N = 143,678
- Reported low English fluency N = 73,813

Final sample: N = 103,575 (71% female)

27 nations: Morocco, Kenya, Lebanon, Namibia, Egypt, Nigeria, South Africa, Canada, United States, Mexico, Jamaica, India, Philippines, Indonesia, Bangladesh, Malaysia, Pakistan, Saudi Arabia, Argentina, Columbia, Brazil, Trinidad and Tobago, Romania, United Kingdom, Italy, Germany, Australia

Analytic Plan

- Exploratory Factor Analysis
  - 2-4 factor solutions, Oblimin rotations

- Item-level analyses

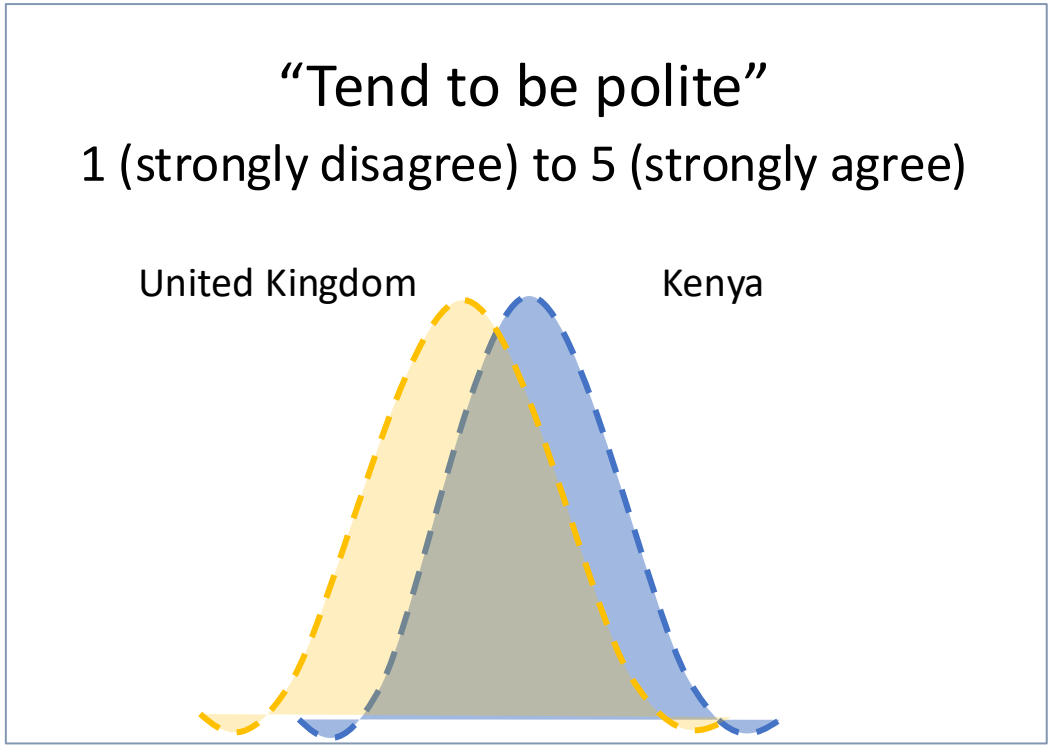
Analytic Plan

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- Item-level analyses
  - Within-session test-retest reliability (items presented ~20 minutes apart)

Analytic Plan

- Exploratory Factor Analysis
  - 2-4 factor solutions, Oblimin rotations

- Item-level analyses
  - Within-session test-retest reliability (items presented ~20 minutes apart)
  - Examine differences between nation pairs in item distributions
    - **13 nations**: United states, Australia, Canada, India, Jamaica, Kenya, Namibia, Nigeria, Pakistan, The Philippines, South Africa, Trinidad and Tobago, Great Britain

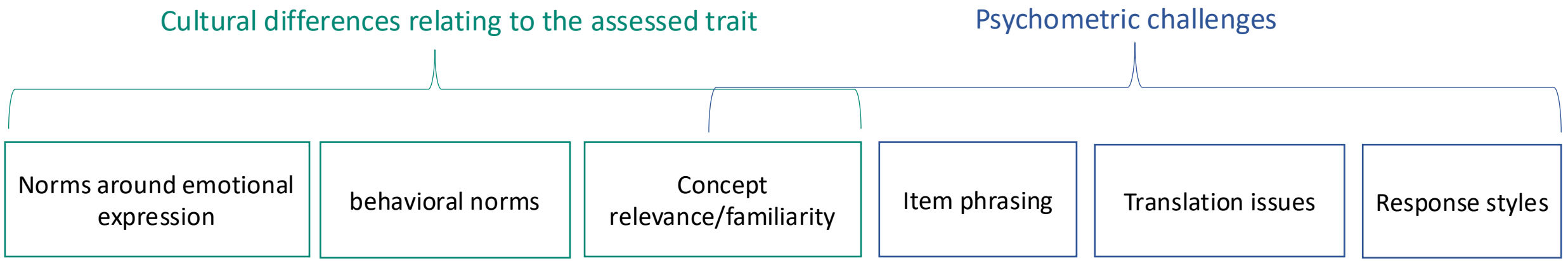


55 items x 78 nation pairs =

4,290 comparisons

Testing whether two samples come from the same population

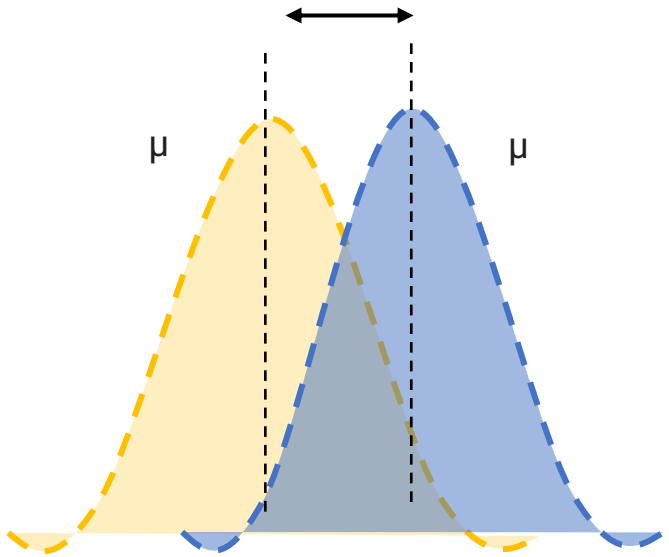
- Identify items that show evidence for generalizability/universality
- Differences can arise for many reasons…..



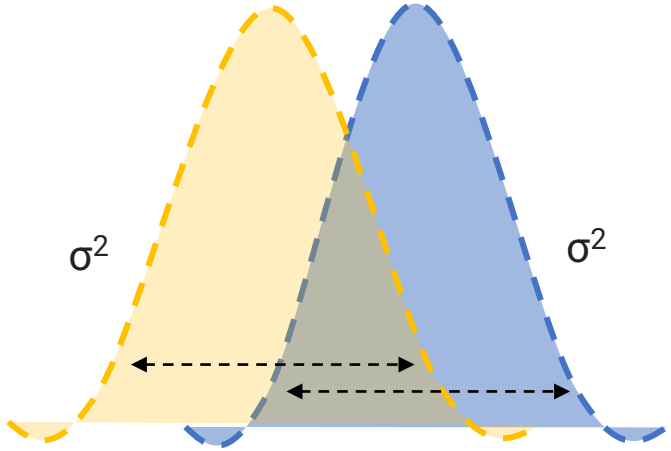
- Goal of present work is to identify items to examine more closely and consider dropping from the final scale

Comparing trait scores across groups

Comparing Means: Test of differences in location (t-test)

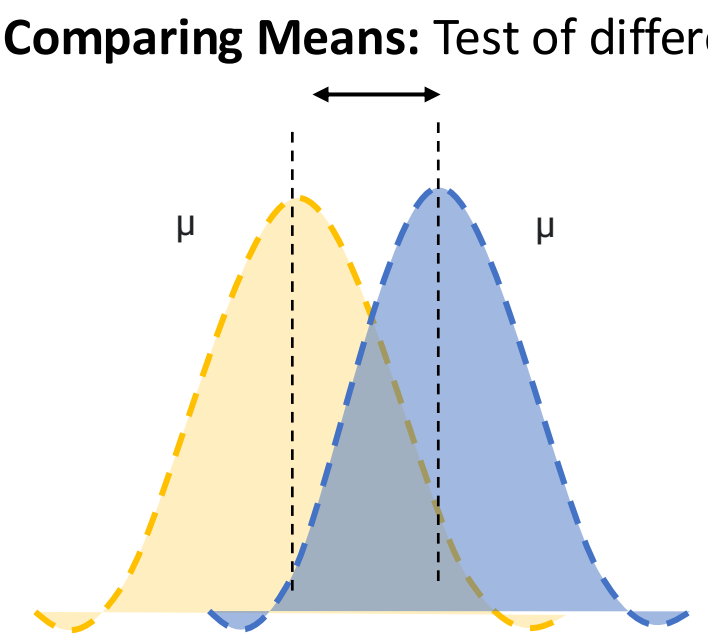


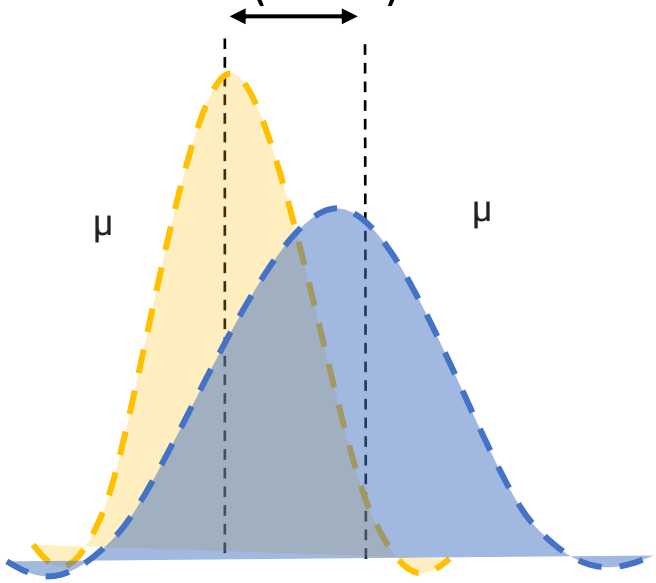
Comparing Variances: Test of differences in spread (F-test)

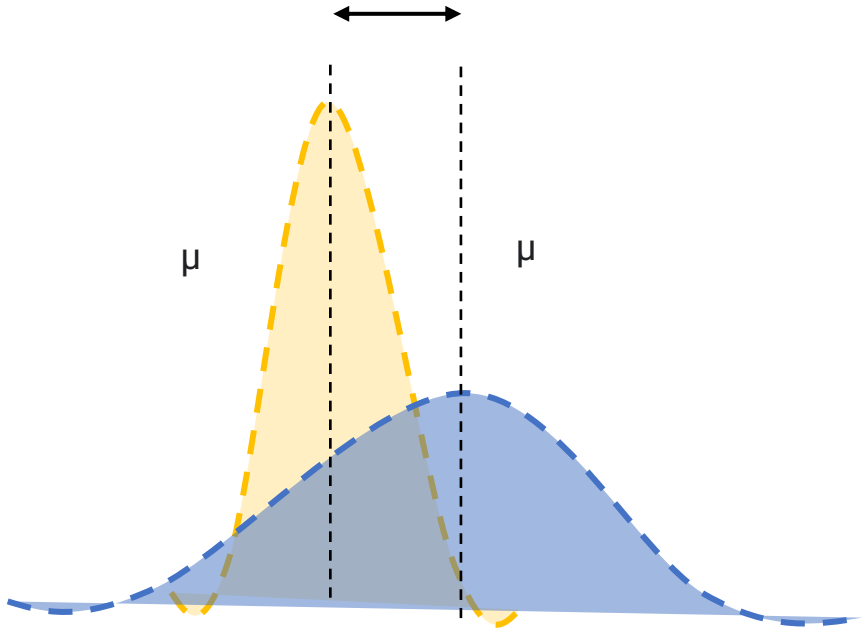


Comparing trait scores across groups

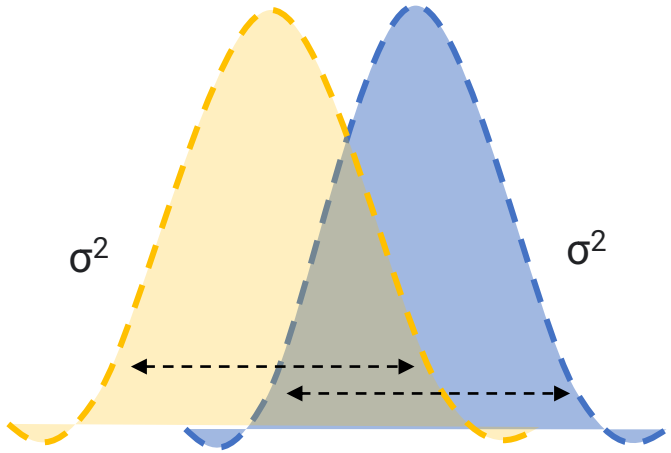
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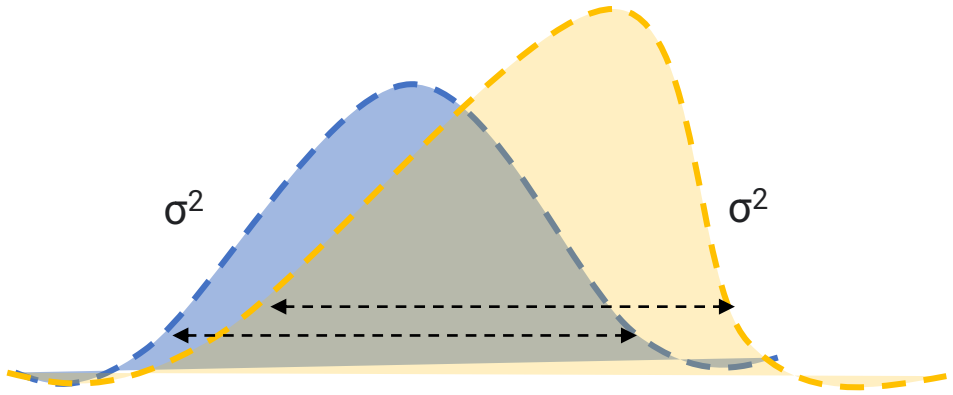


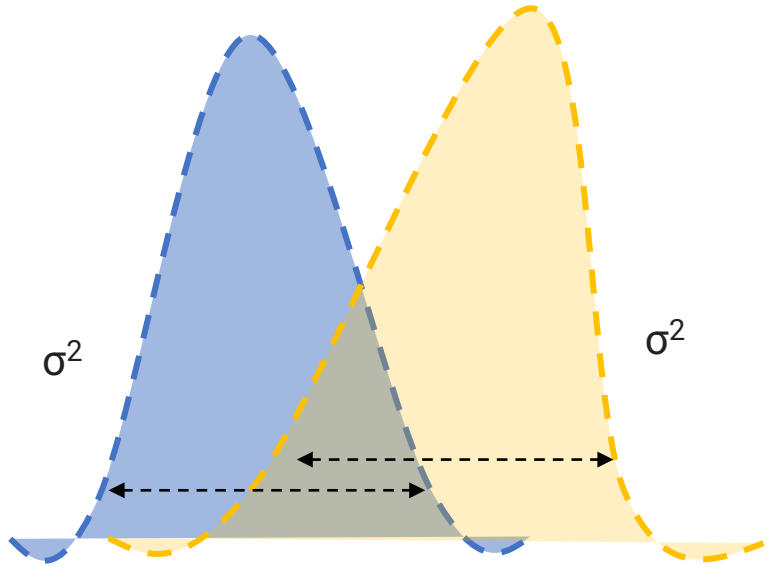


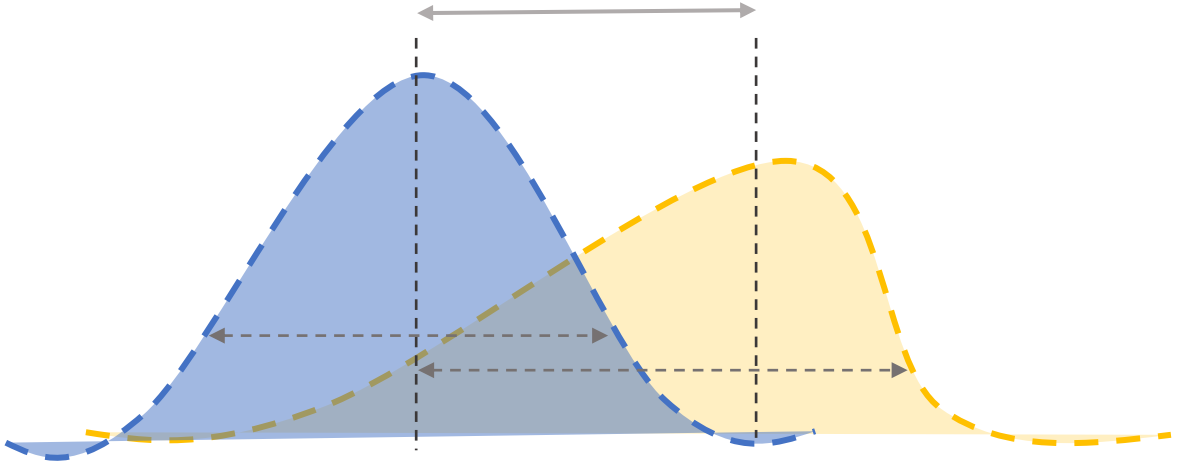


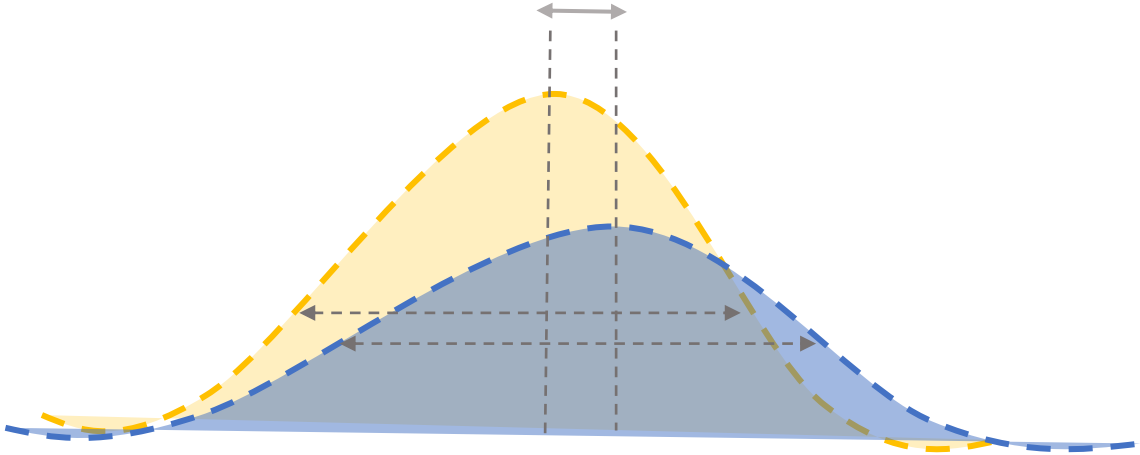
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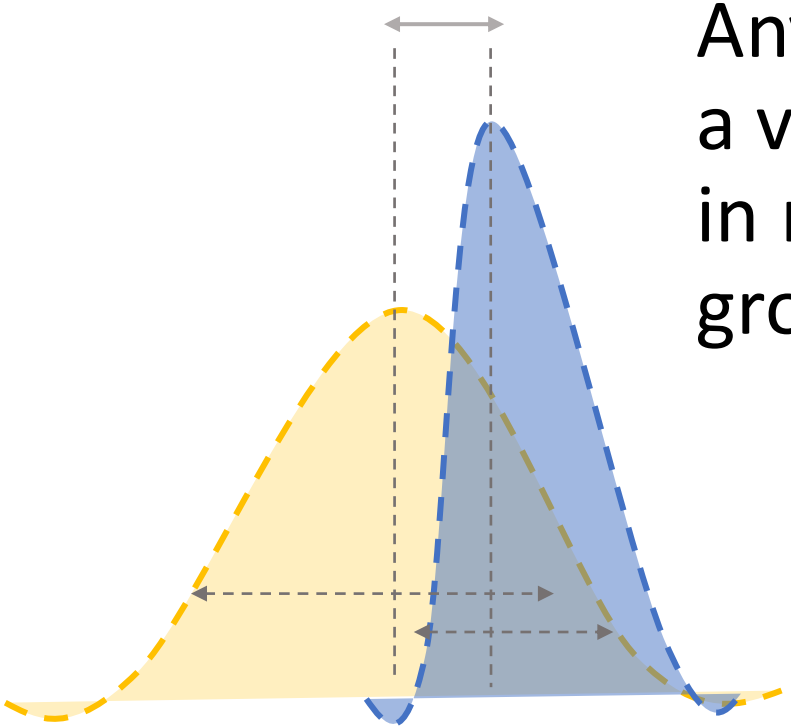




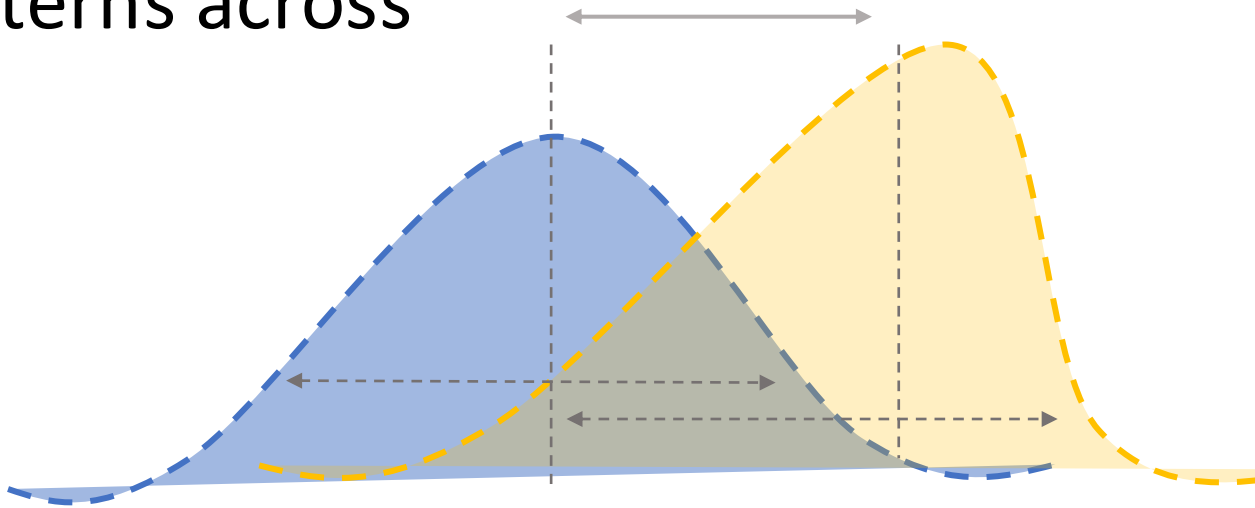


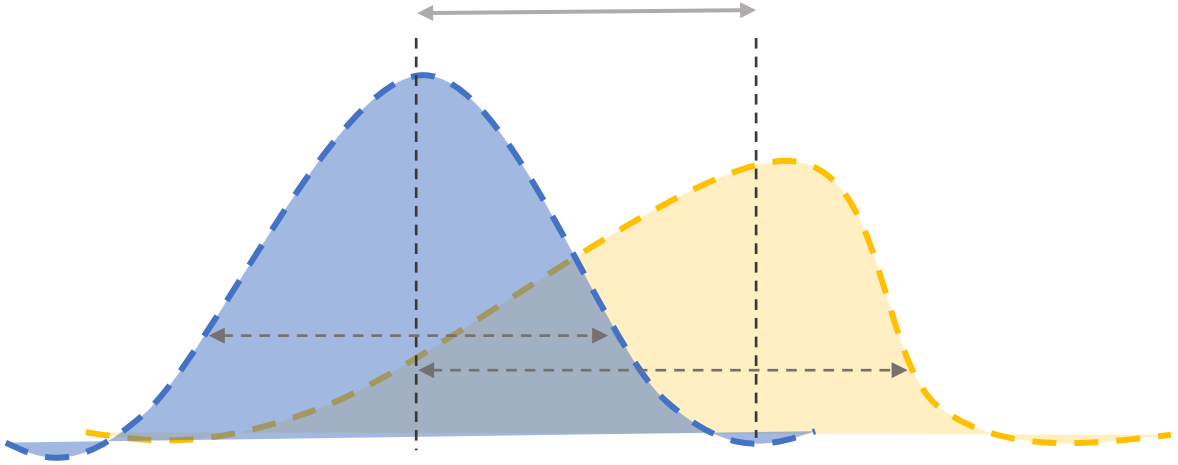


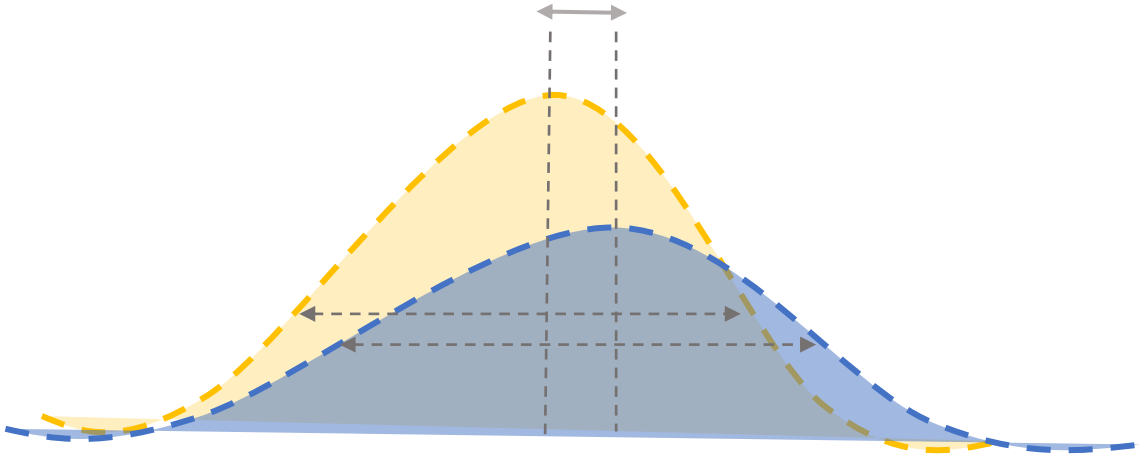


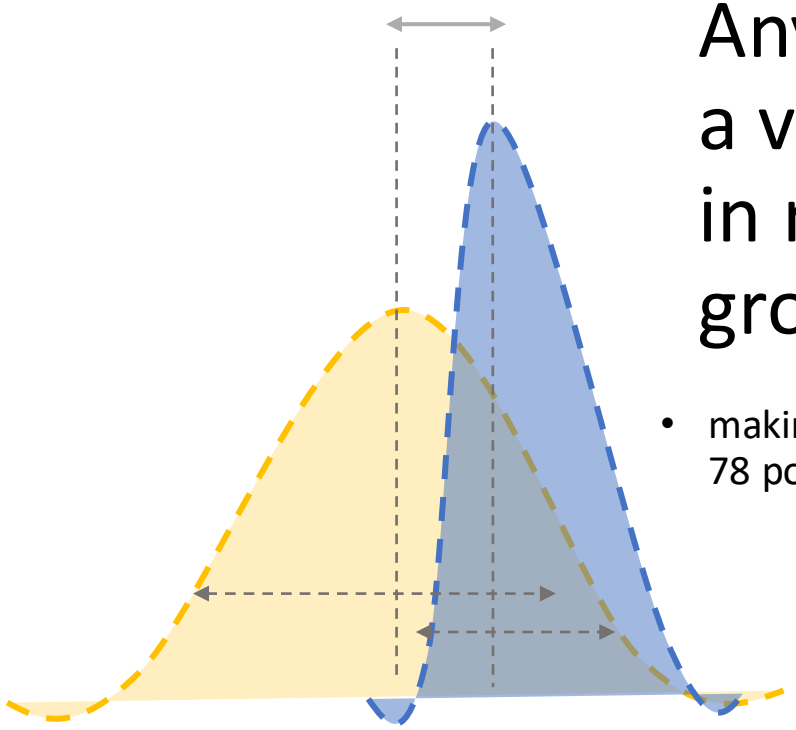


Any single test might fail to detect a variety of important differences in responding patterns across groups.



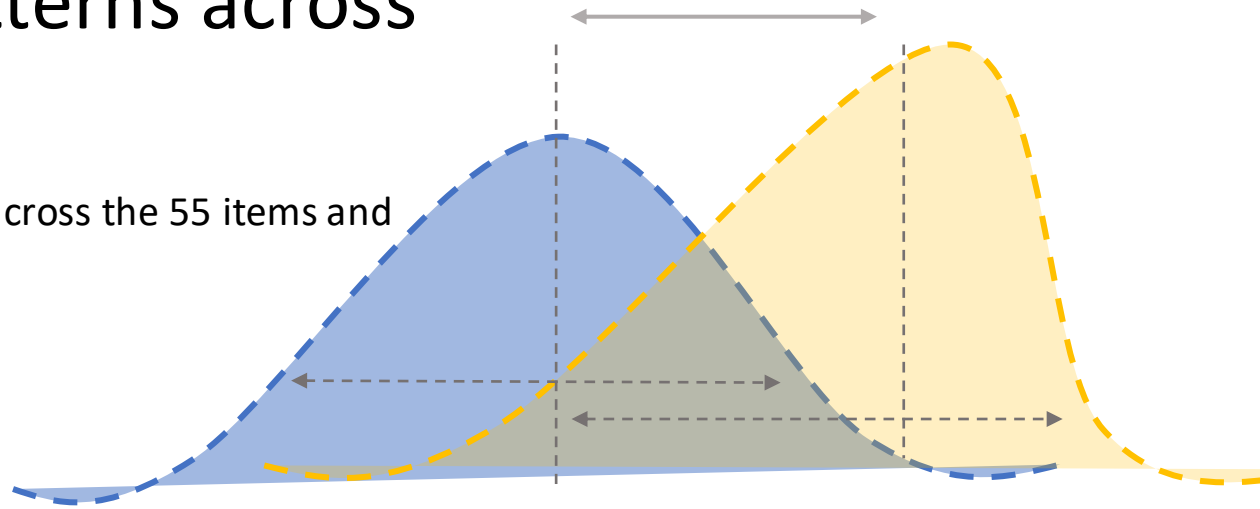






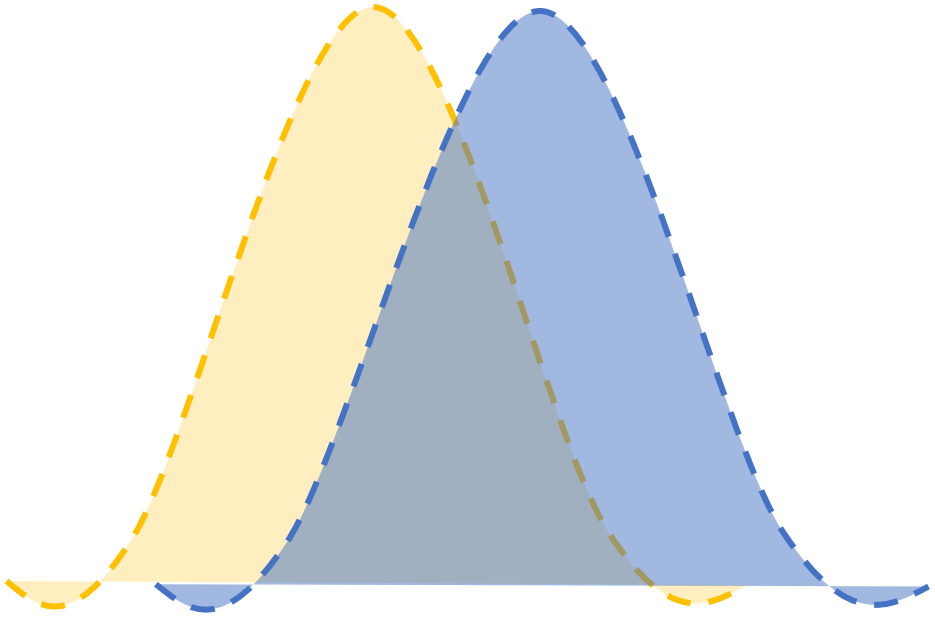
Any single test might fail to detect a variety of important differences in responding patterns across groups.

- making **4,290** total comparisons (across the 55 items and 78 possible country pairs)

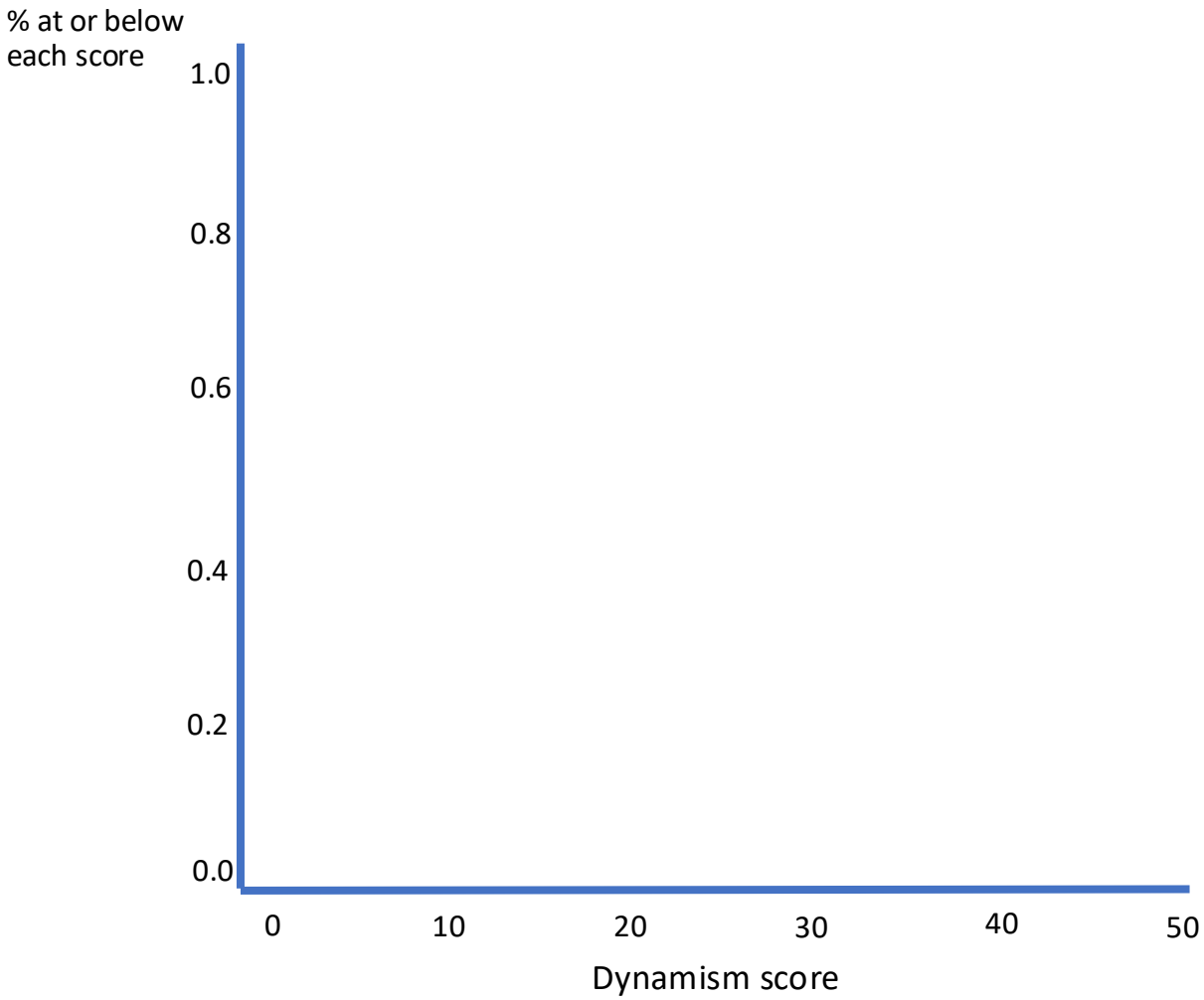


Comparing entire distributions using a single test

Distribution of Dynamism scores for two samples

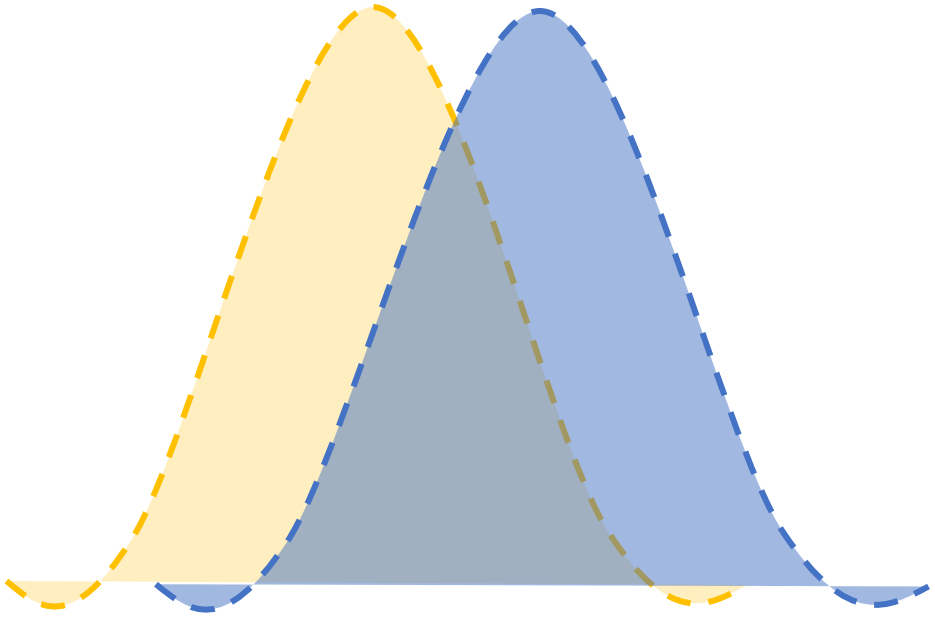


Empirical cumulative distribution function for each group

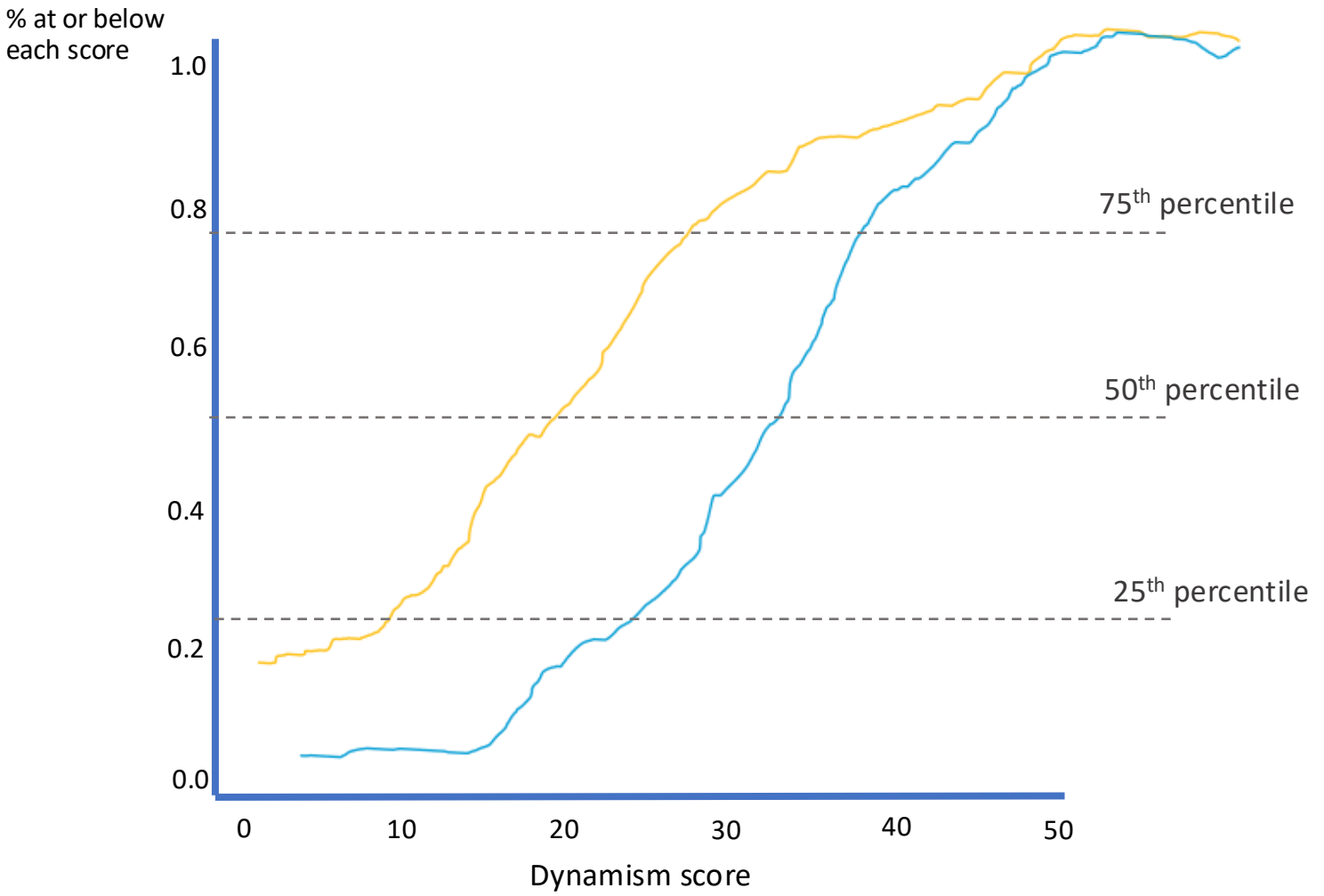


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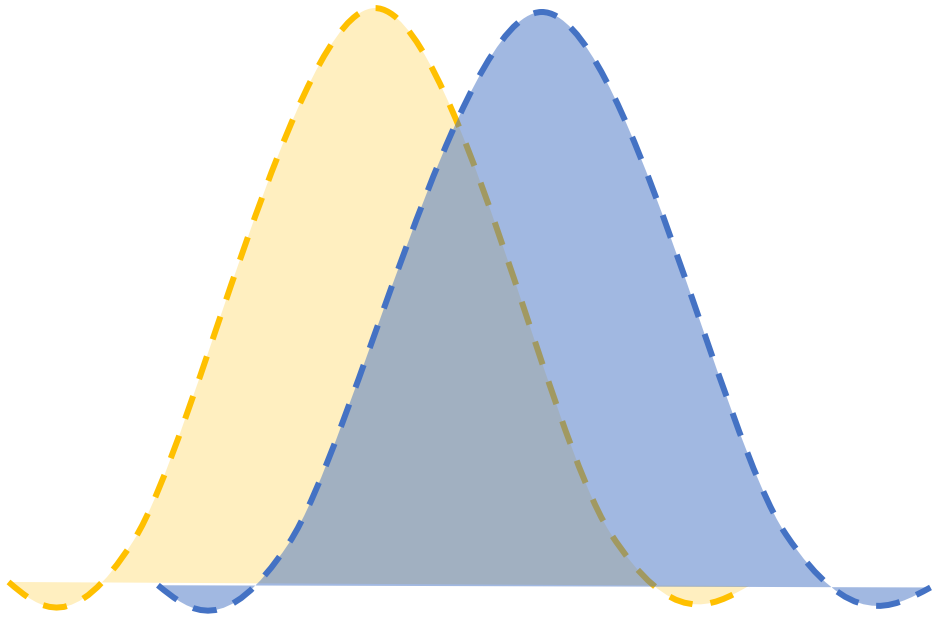


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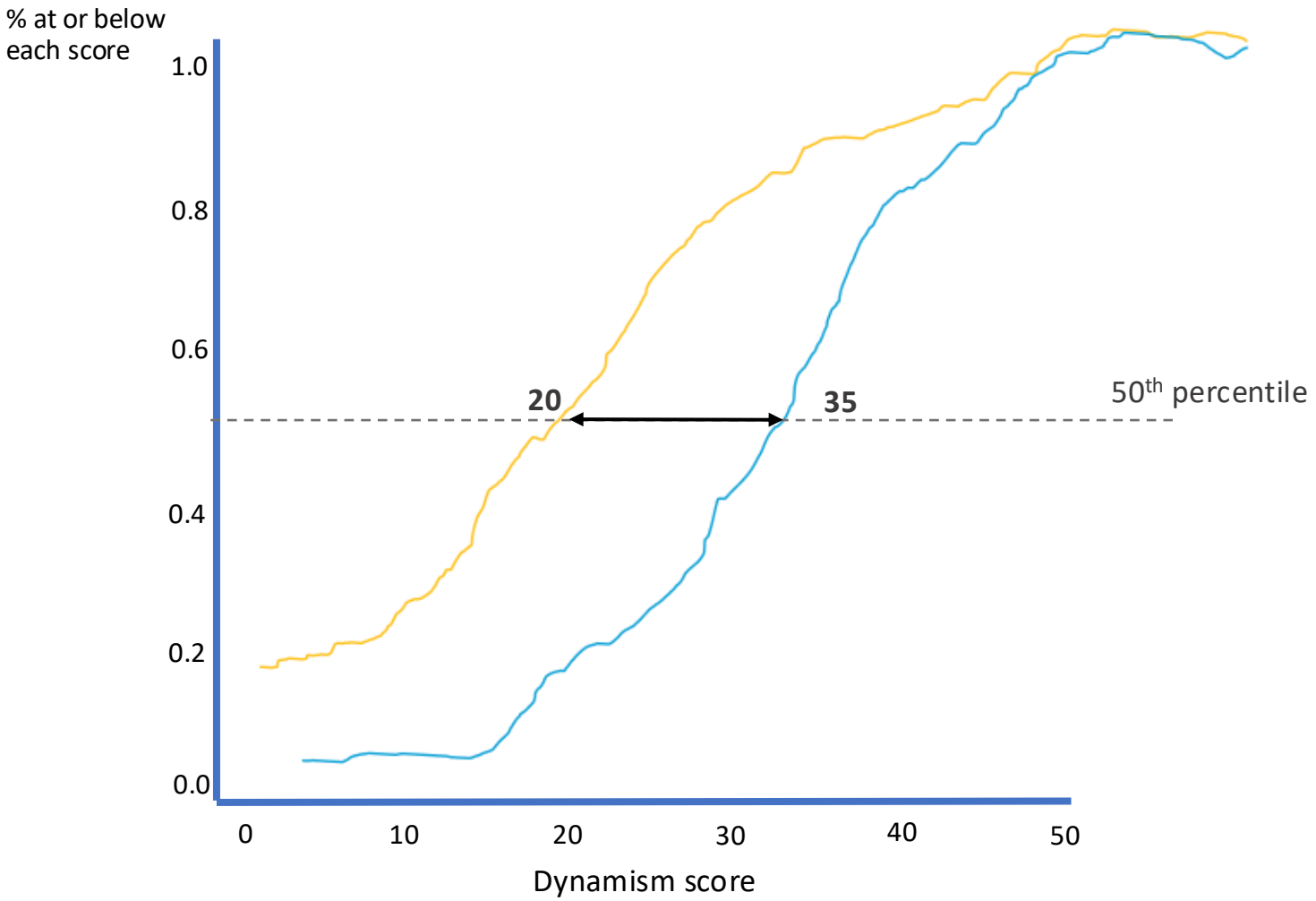


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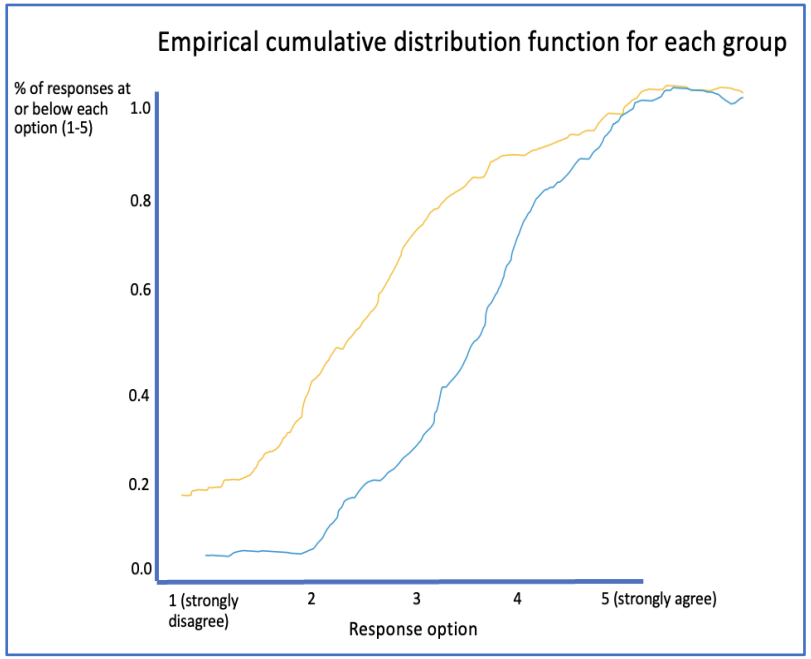


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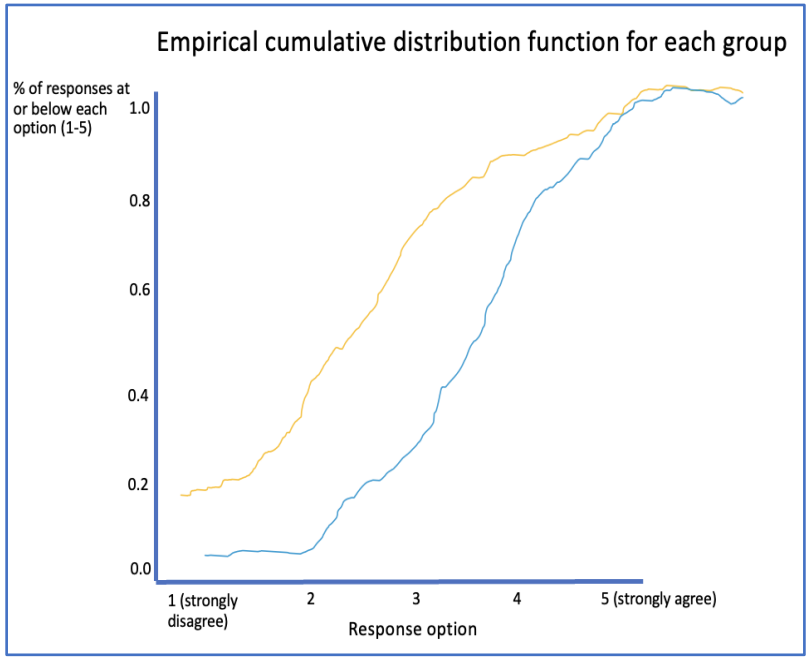


Comparing entire distributions using a single test: Several statistical options for comparing ECDFs

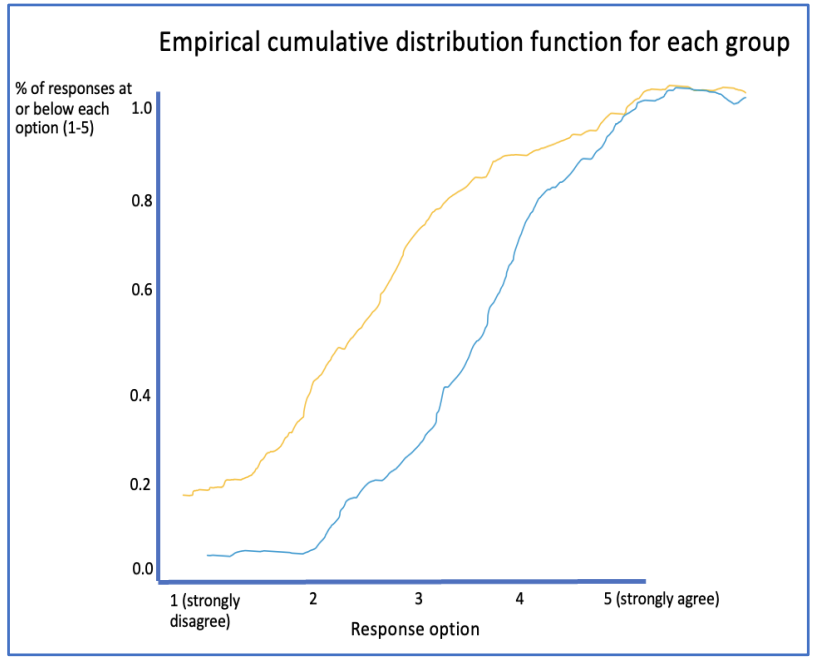
Anderson-Darling (AD) statistic



Wasserstein statistic

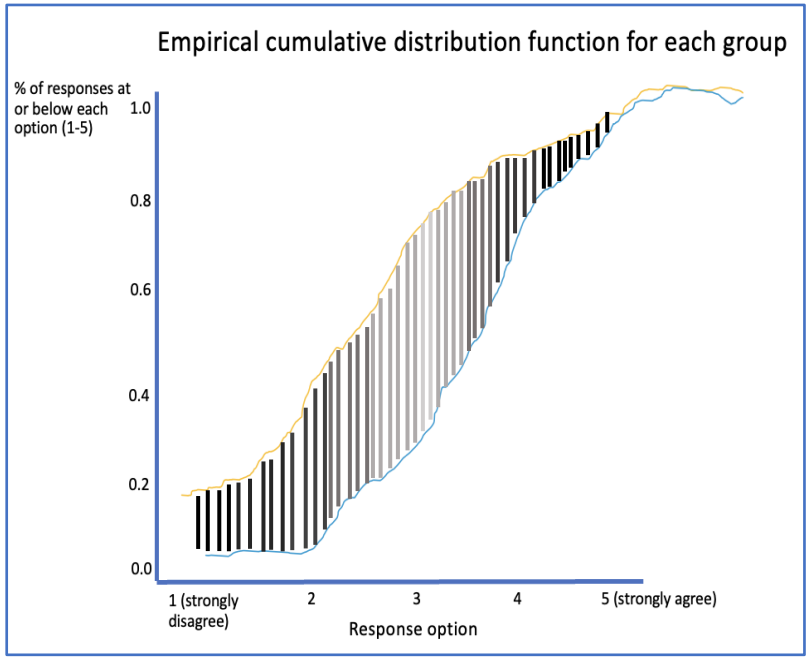


DTS statistic

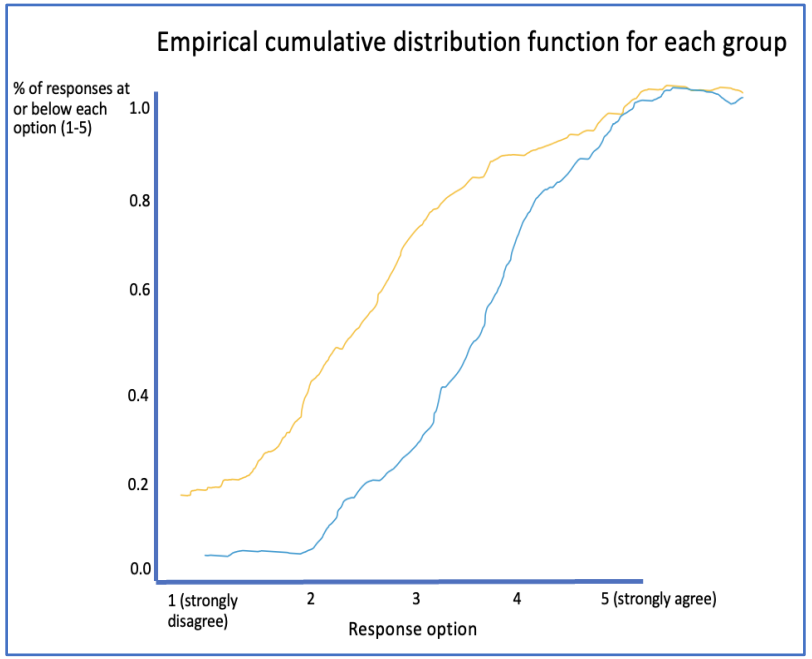


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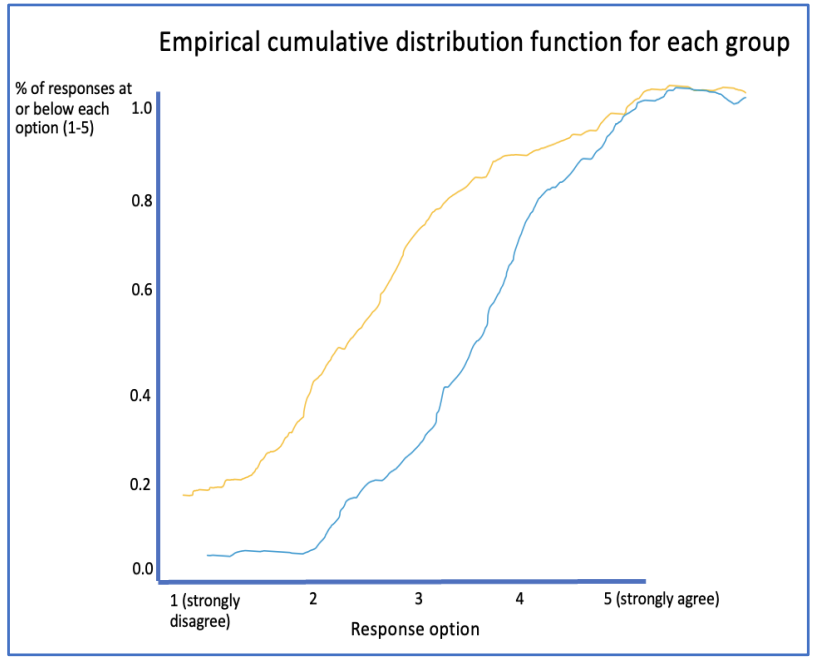
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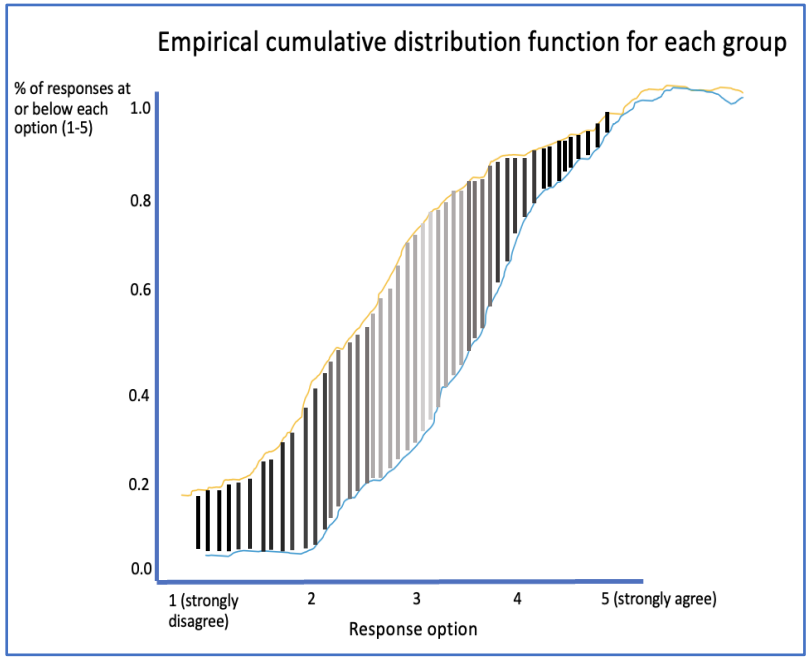


The weighted sum of all observed differences (Anderson & Darling, 1952)

Putting less weight on observations in the middle of the distribution is done to improve the power of the test.

Comparing entire distributions using a single test: Several statistical options for comparing ECDFs

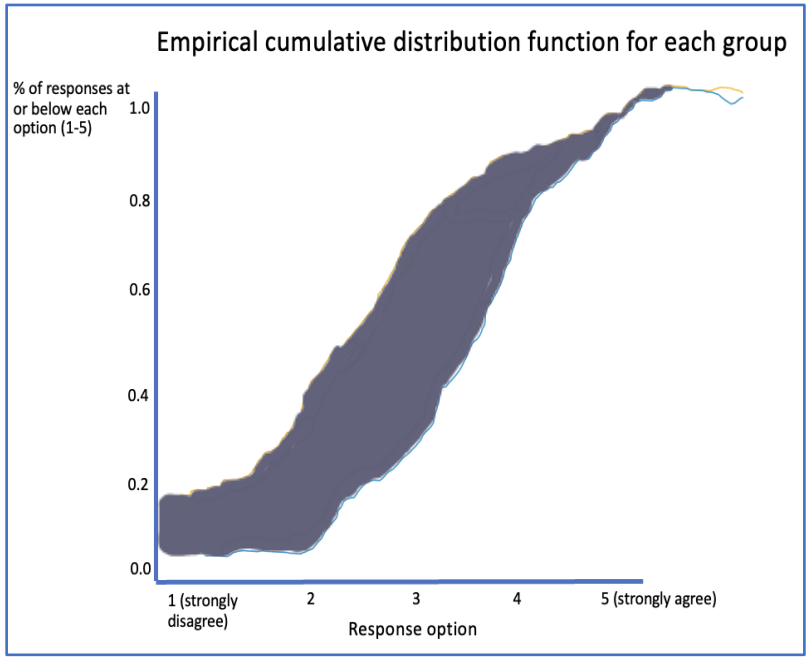
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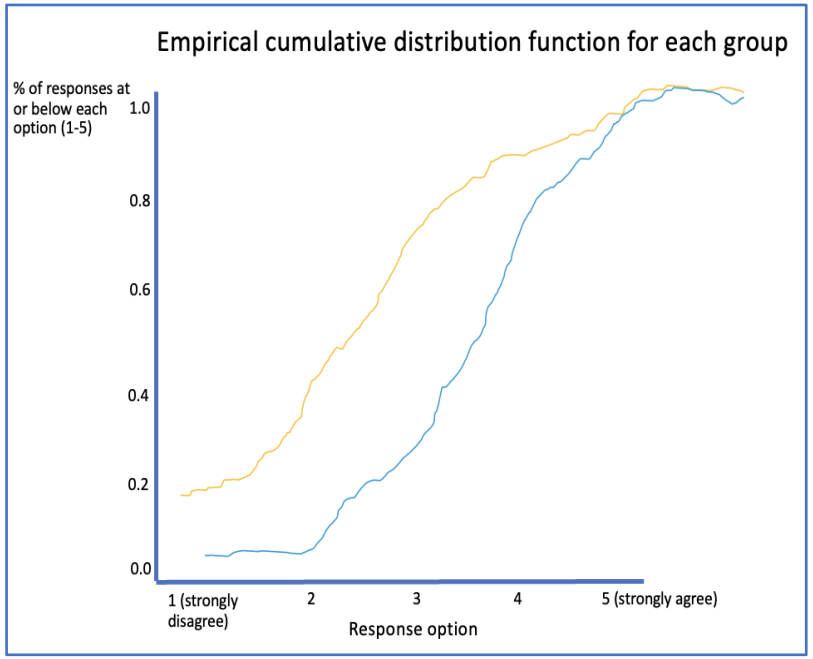
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Wasserstein statistic



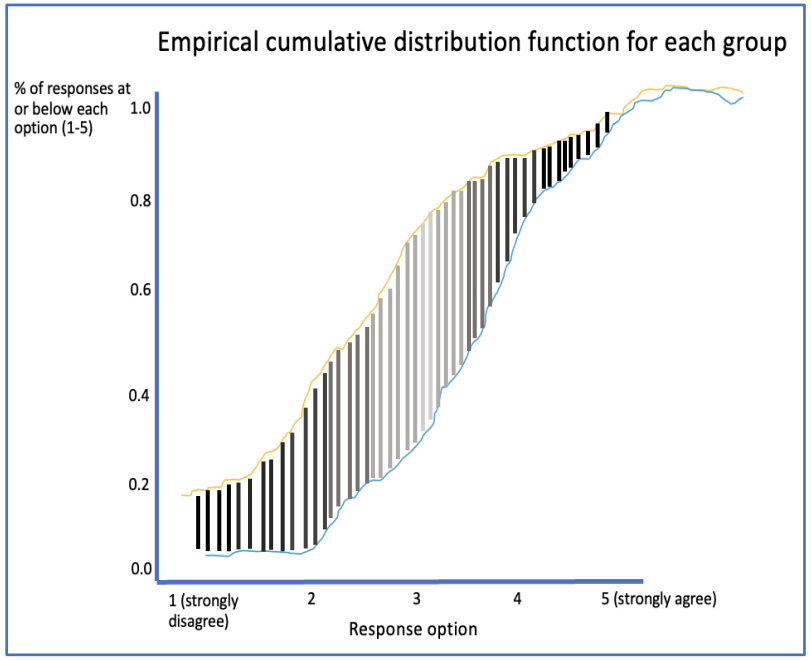
The area between the ECDFs (Wasserstein, 1969)

DTS statistic



Comparing entire distributions using a single test: Several statistical options for comparing ECDFs

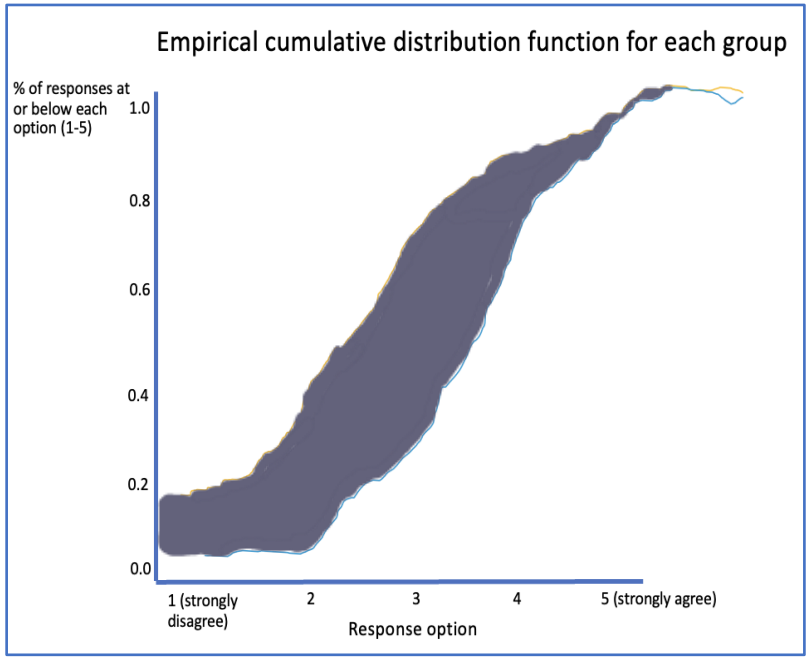
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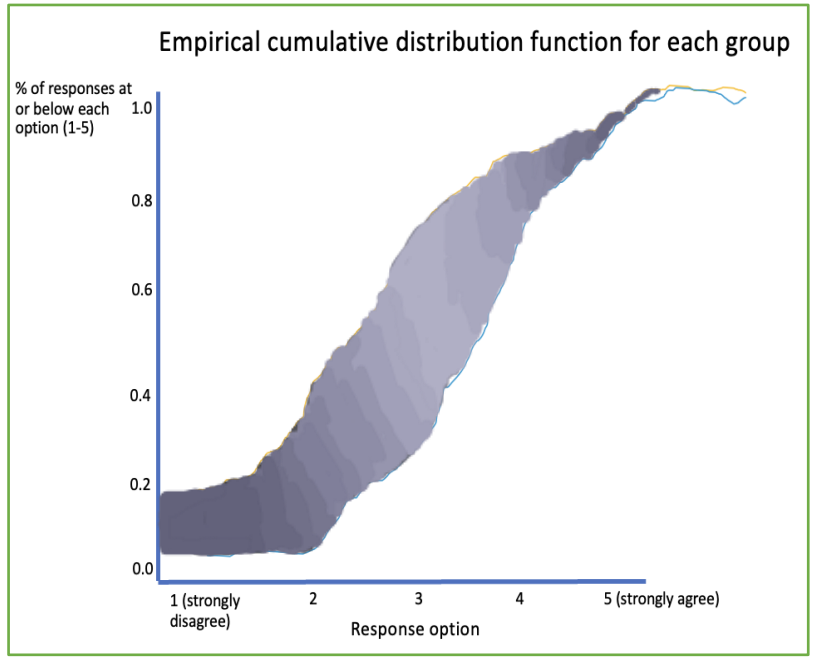
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Wasserstein statistic



The area between the ECDFs (Wasserstein, 1969)

DTS statistic ☑

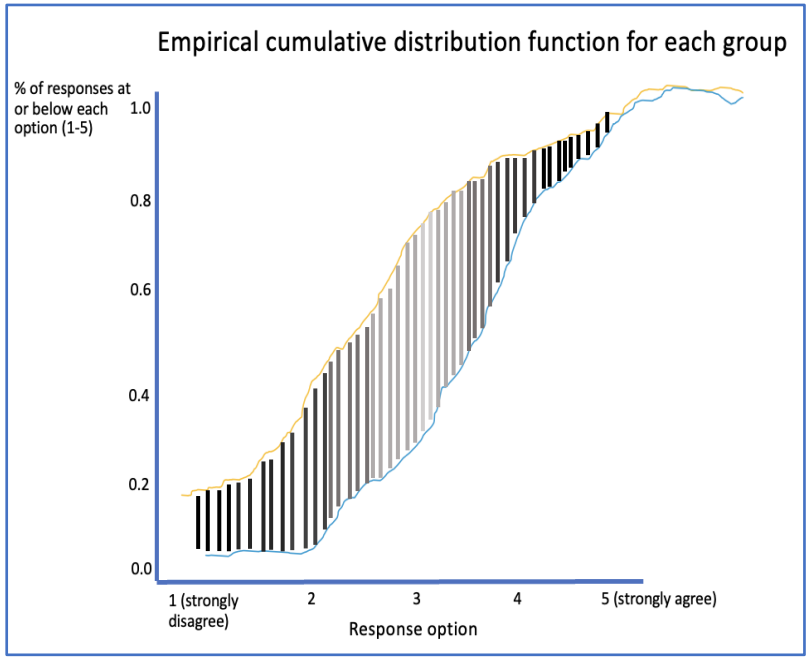


Synthesis of AD and Wasserstein: The weighted integral of the difference between the ECDFs (Dowd, 2020)

Comparing entire distributions using a single test: Several statistical options for comparing ECDFs

Significant p-value (*p*<.05) indicates that the difference between the ECDFs is fairly extreme if we assume that that samples are drawn from the same population

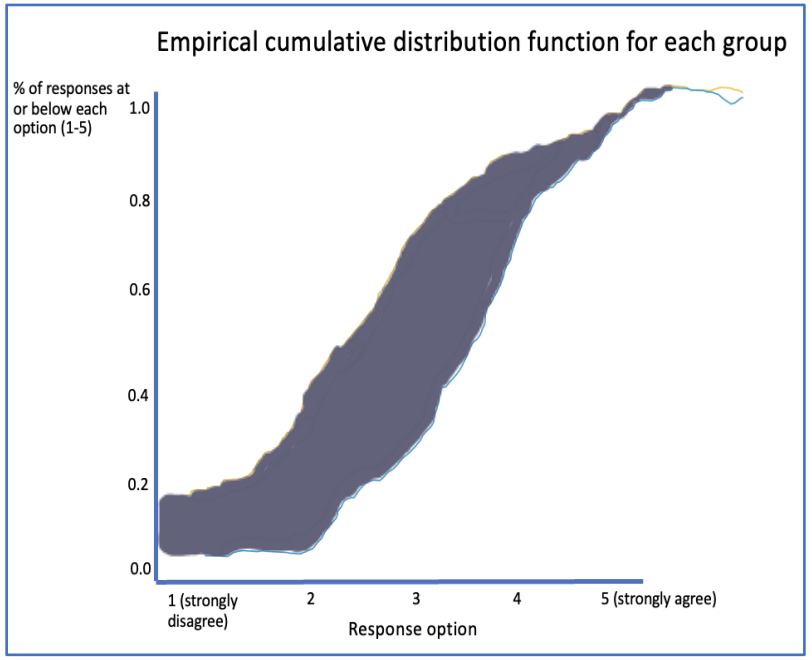
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The weighted sum of all observed differences (Anderson & Darling, 1952)

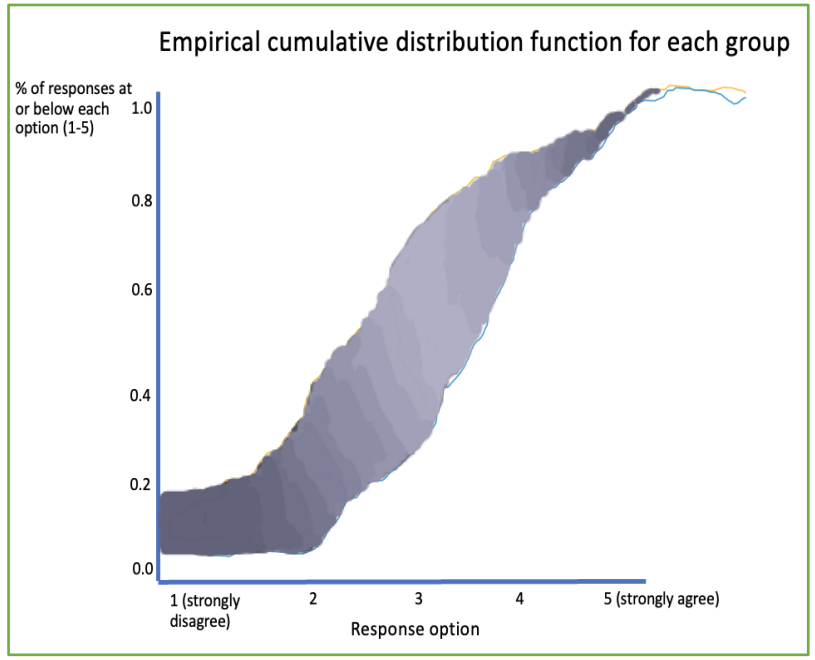
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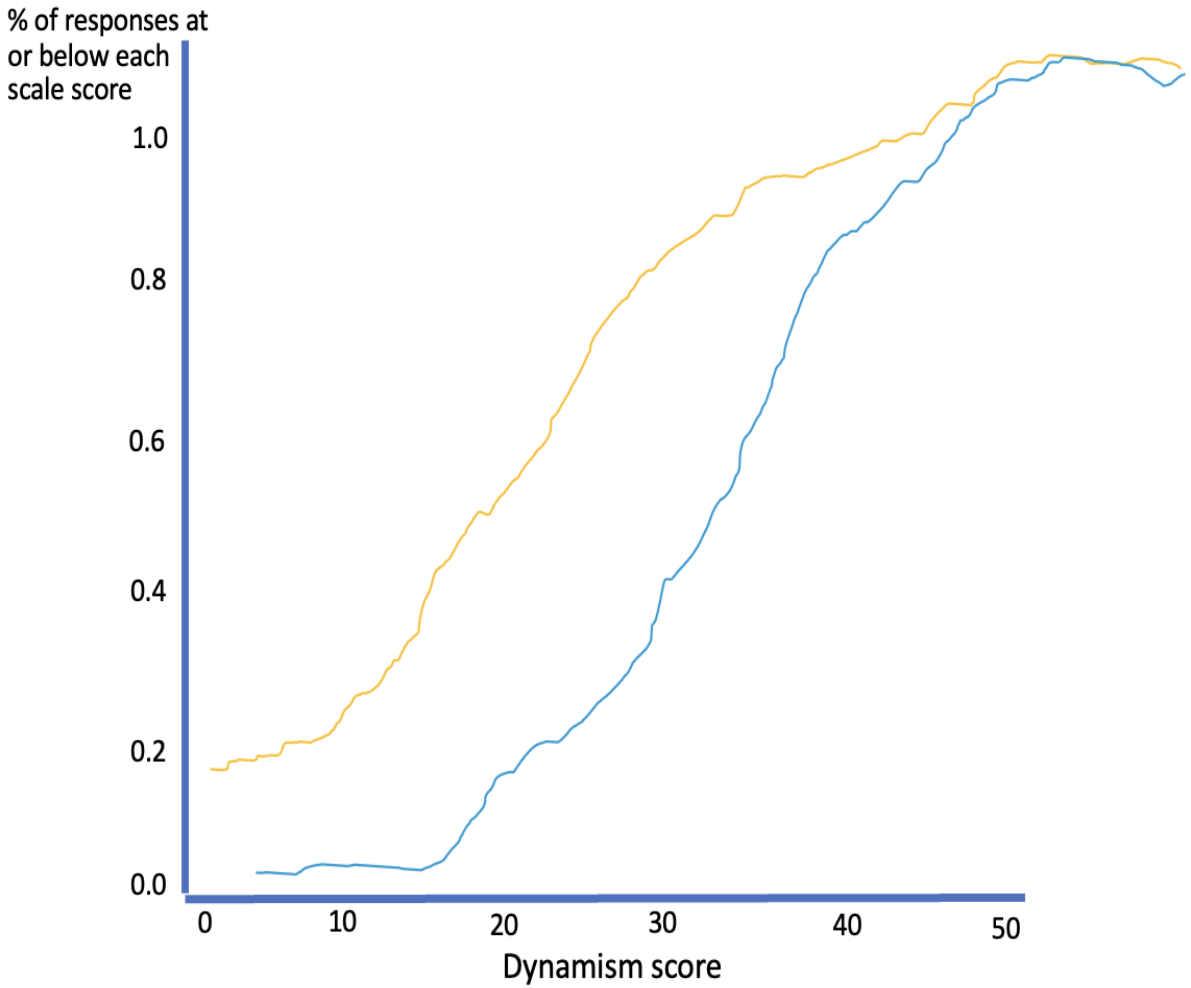


Synthesis of AD and Wasserstein: The weighted integral of the difference between the ECDFs (Dowd, 2020)

ECDFs for scale scores vs. item scores

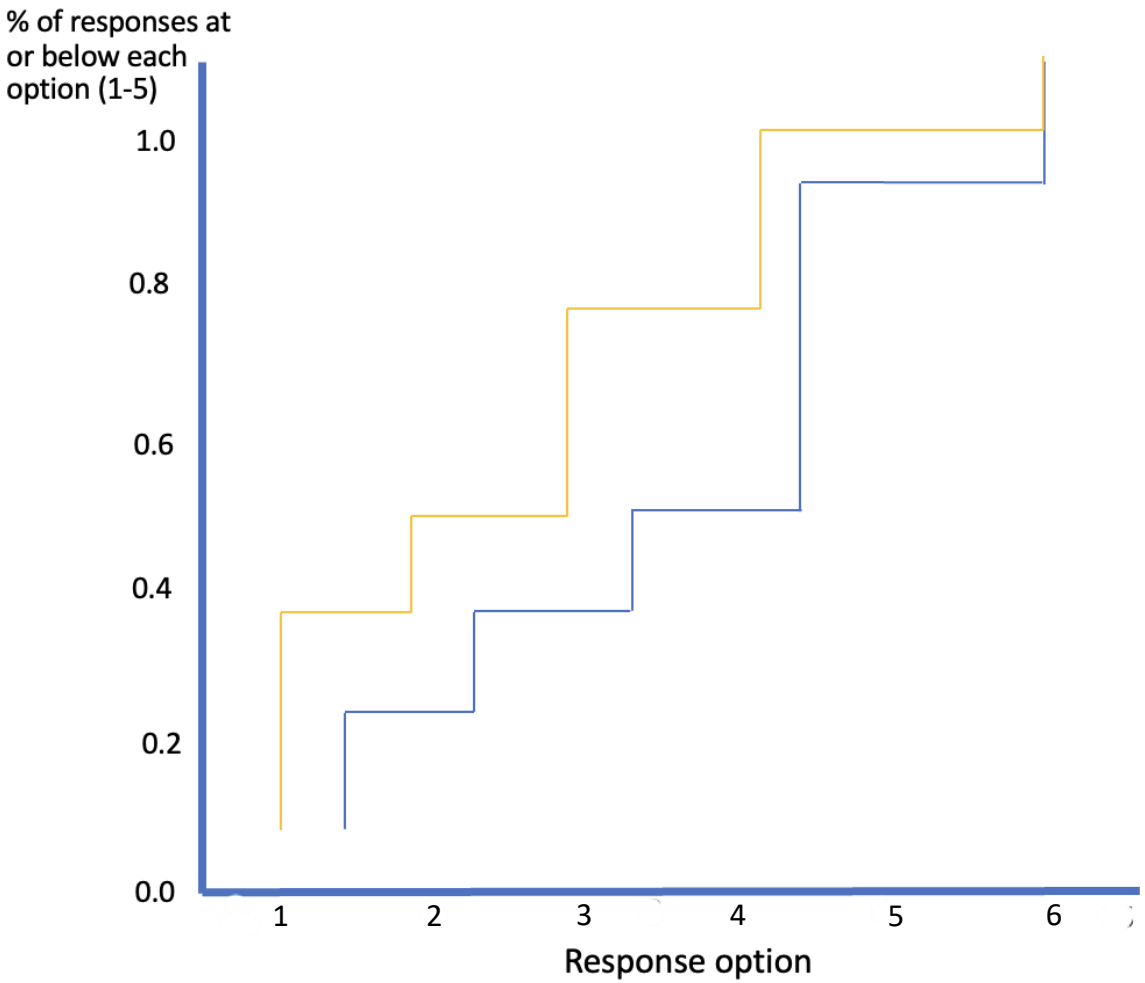
Empirical cumulative distribution function for each group

Dynamism scale score



Empirical cumulative distribution function for each group

Dynamism item: "Tend to be confident"



Results:

Item-level analyses	Factor analyses	Dropping items

Results:

Item-level analyses	**Factor analyses**	**Dropping items**

71% of DTS tests were significant overall

55 (# items) x 78 (nation pairs) = **4,290 tests**

13 nations:

United states
Australia
Canada
India
Jamaica
Kenya
Namibia
Nigeria
Pakistan
The Philippines
South Africa
Trinidad and Tobago
Great Britain

Results:

Item-level analyses | **Factor analyses** | **Dropping items**

71% of DTS tests were significant overall

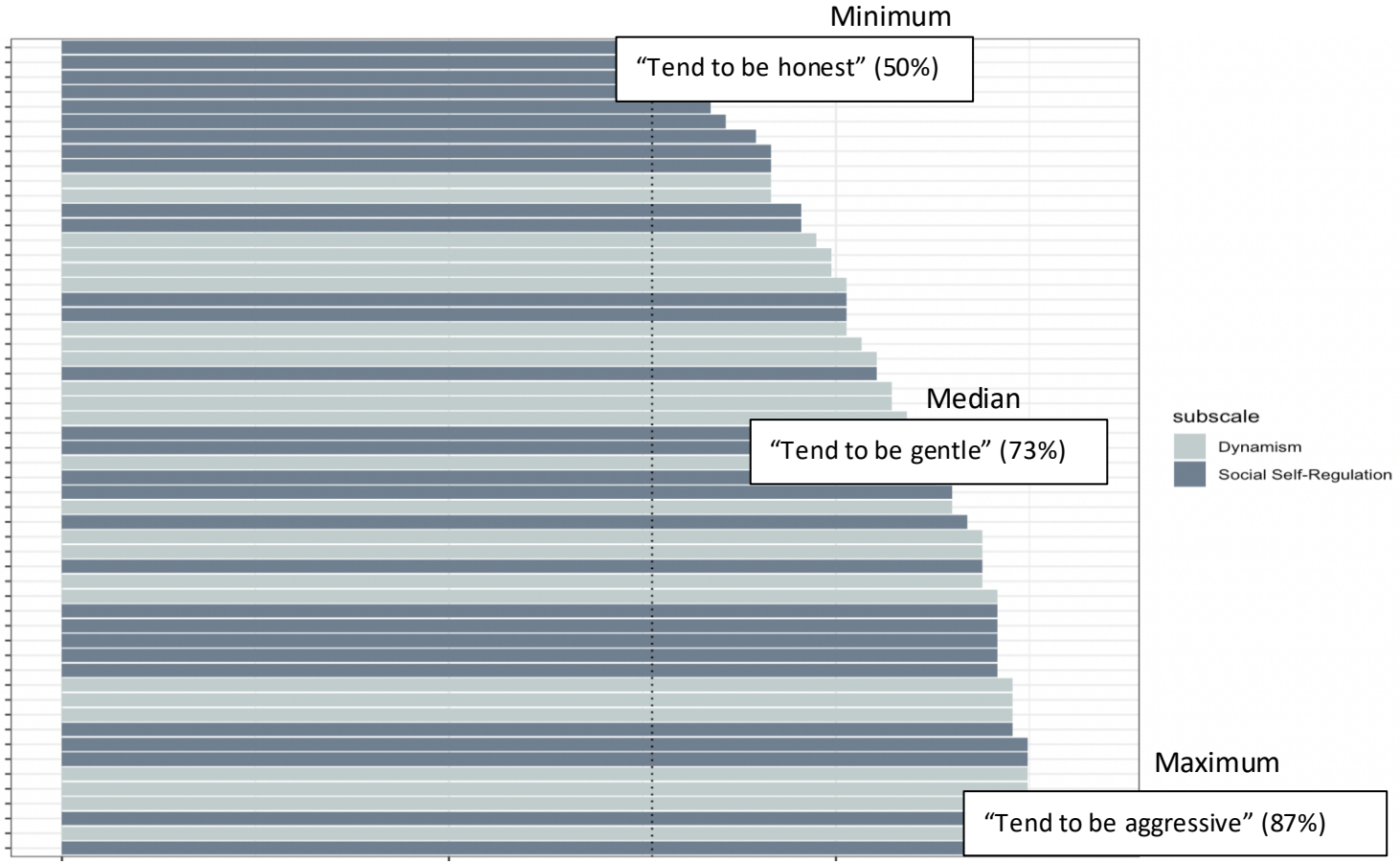
55 (# items) x 78 (nation pairs) = **4,290 tests**

Items with the highest and lowest number of significant tests

-"Tend to be aggressive"
-"Tend to be silent"
-"Tend to be naïve"
-"Tend to be shy"
-"Tend to be a loner"
-"Tend to be bashful"

-"Tend to be honest"
-"Tend to be respectful"
-"Tend to be humane"
-"Tend to be compassionate"
-"Tend to be kind"
-"Tend to be good-natured"

Proportion of tests significant for each item



Results:

Item-level analyses	**Factor analyses**	**Dropping items**

71% of DTS tests were significant overall

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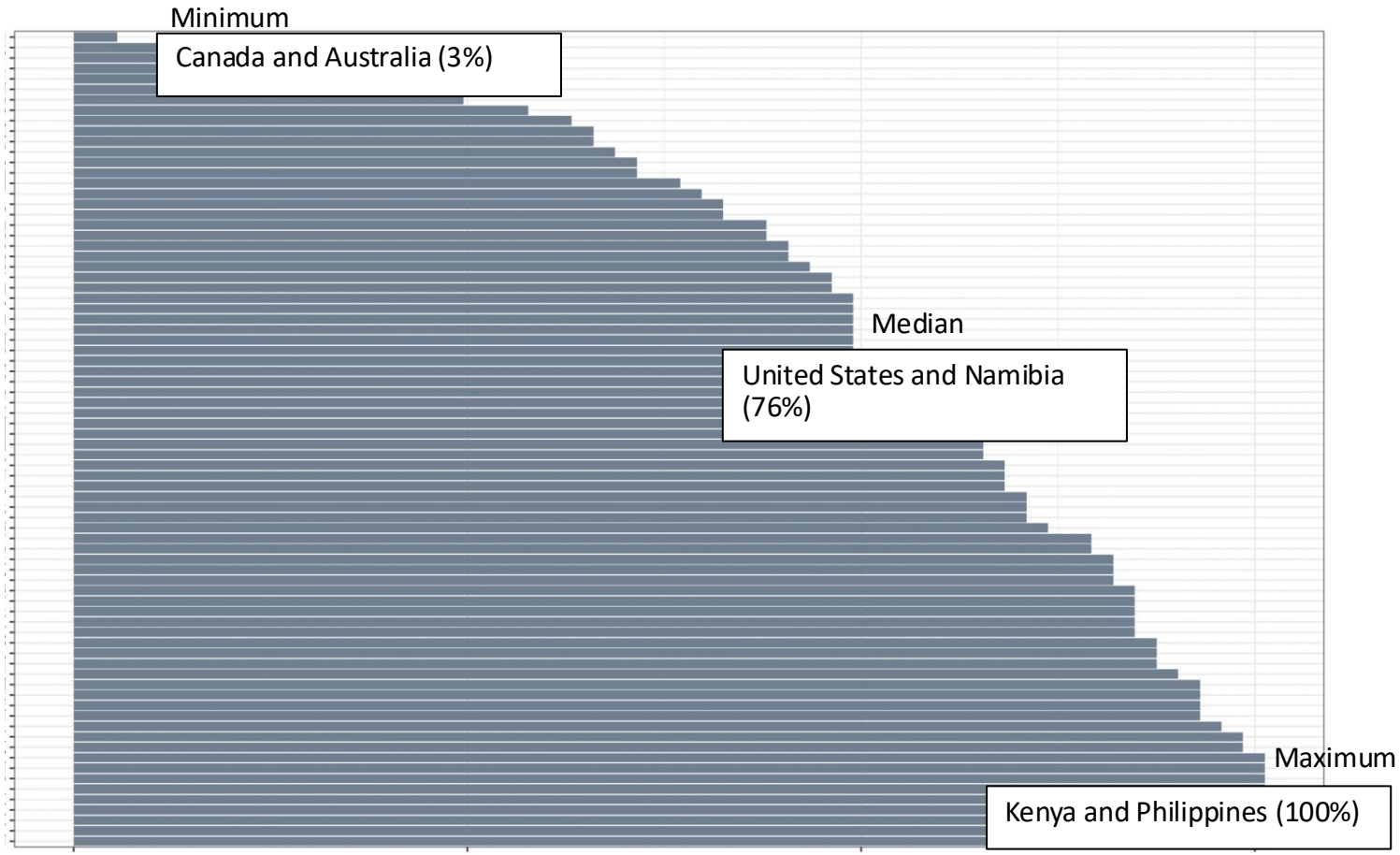
Nation pairs with the highest and lowest number of significant tests

-Kenya-Philippines
-Kenya-United Kingdom
-Kenya-India
-Kenya-United States
-United States-Pakistan
-United States-India
-Philippines-Nigeria
-Philippines-Pakistan
-Philippines-India

All pairs of….
-Australia/Canada/United Kingdom/United States

-Namibia-South Africa
-Namibia-Nigeria
-Namibia-Trinidad and Tobago
-Jamaica-Trinidad and Tobago

Proportion of tests significant for each nation pair



Results:

Item-level analyses | **Factor analyses** | **Dropping items**

71% of DTS tests were significant overall

55 (# items) x 78 (nation pairs) = **4,290 tests**

Nation pairs with the highest and lowest number of significant tests

-Kenya-Philippines
-Kenya-United Kingdom
-Kenya-India
-Kenya-United States
-United States-Pakistan
-United States-India
-Philippines-Nigeria
-Philippines-Pakistan
-Philippines-India

All pairs of….
-Australia/Canada/United Kingdom/United States

-Namibia-South Africa
-Namibia-Nigeria
-Namibia-Trinidad and Tobago
-Jamaica-Trinidad and Tobago

Percentage of significant tests per individual nation

Nation	% significant
Philippines	91.4%
Kenya	85.6%
India	80.5%
Nigeria	76.8%
Pakistan	76.8%
USA	77.4%
Jamaica	71.1%
United Kingdom	71.8%
South Africa	61.8%
Trinidad and Tobago	64.4%
Canada	61.4%
Namibia	55.6%
Australia	54.2%

Results:

Item-level analyses	**Factor analyses**	**Dropping items**

Within-session test-retest reliability

Results:

Item-level analyses	**Factor analyses**	**Dropping items**

Within-session test-retest reliability

Highest test-retest

Tend to be…

“gossipy” (r = .79)

“quiet” (r = .79)

“sad” (r = .78)

“a loner” (r = .78)

“aggressive” (r = .78)

“hot-tempered” (r = .76)

Lowest test-retest

Tend to be…

“courteous” (r = .50)

“humane” (r = .54)

“respectful” (r = .55)

“kind” (r = .56)

“faithful” (r = .56)

“polite” (r = .56)

High and low item stability appear to be related to social desirability (or undesirability)

Results:

Item-level analyses	Factor analyses	Dropping items

Results:

Item-level analyses | **Factor analyses** | **Dropping items**

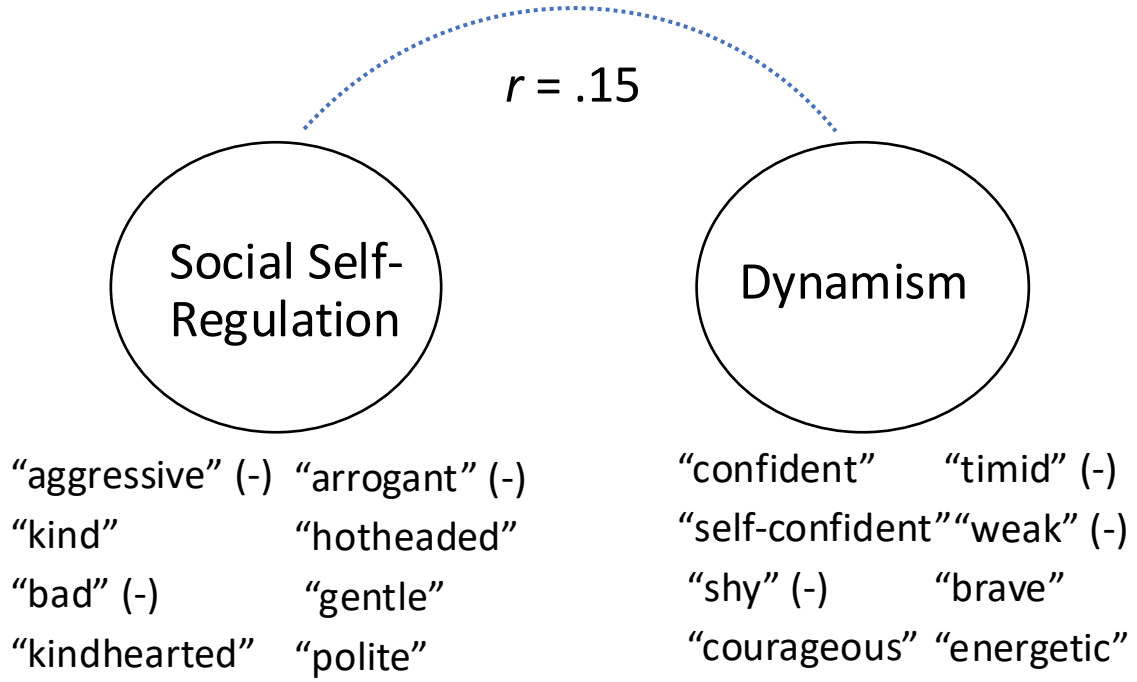
Fit indices

- Ipsatized data, oblimin rotations

	RMSR	TLI	RMSEA	BIC
Two factors	.07	.281	.144	2933961
Three factors	.05	.308	.141	2716522
Four factors	.04	.315	.140	2584020

- Only a small improvement in fit indices when extracting 3 or 4 factors instead of 2

2 factor solution



"naïve", "hardworking", "industrious" had primary loadings onto dynamism instead of social self-regulation

Results:

Item-level analyses	Factor analyses	Dropping items

Results:

Item-level analyses	Factor analyses	Dropping items

Items to look more closely at and consider dropping….

- “Hardworking”, “Industrious”, “Naïve”—primary loadings onto unintended construct, and “naïve” also had a high number of significant DTS tests

High number of DTS tests, high item stability

- “A loner”
- “Aggressive”

Low number of DTS tests, Low item stability

- “Respectful”
- “kind”
- “humane”

Ongoing qualitative work suggests we should drop other items (e.g., “bashful”—which also had a high number of significant DTS tests here) because they are not well understood in many nations.

Results:

Item-level analyses	Factor analyses	Dropping items

Takeaways

- Results point to a few items to consider dropping
  - Lots of variation in responding, possible social desirability issues, etc.
- Many differences in responding at the item-level across nations
- Results influenced by the specific set of nations included
- Findings from this work and the qualitative work will be compared before dropping items

Results:

Item-level analyses	Factor analyses	Dropping items

Takeaways

- Results point to a few items to consider dropping
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Percentage of significant tests per individual nation

Nation	% significant
Philippines	91.4%
Kenya	85.6%
India	80.5%
Nigeria	76.8%
Pakistan	76.8%
USA	77.4%
Jamaica	71.1%
United Kingdom	71.8%
South Africa	61.8%
Trinidad and Tobago	64.4%
Canada	61.4%
Namibia	55.6%
Australia	54.2%

Results:

Item-level analyses	Factor analyses	Dropping items

Takeaways

- Results point to a few items to consider dropping
  - Lots of variation in responding, possible social desirability issues, etc.
- Many differences in responding at the item-level across nations
- Results influenced by the specific set of nations included
- Findings from this work and the qualitative work will be compared before dropping items

Percentage of significant tests per individual nation

Nation	% significant	
Philippines	91.4%	(65.5% with Australia)
Kenya	85.6%	(54.5% with Nigeria)
India	80.5%	(61.8% with Trinidad and Tobago)
Nigeria	76.8%	(41.8% with Namibia)
Pakistan	76.8%	(43.6% with Namibia)
USA	77.4%	(21.8% with Namibia)
Jamaica	71.1%	(32.7% with Trinidad and Tobago)
United Kingdom	71.8%	(16.4% with Australia)
South Africa	61.8%	(20% with Namibia)
Trinidad and Tobago	64.4%	(32.7% with Jamaica)
Canada	61.4%	(3.6% with Australia)
Namibia	55.6%	(20% with South Africa)
Australia	54.2%	(3.6% with Canada)

Questions?

Results within each nation: Australia

Nation	ISO3	number_significant	proportion
Canada	CAN	2	3.6 %
United Kingdom	GBR	9	16.4 %
USA	USA	12	21.8 %
South Africa	ZAF	25	45.5 %
Namibia	NAM	28	50.9 %
Trinidad and Tobago	TTO	32	58.2 %
India	IND	36	65.5 %
Jamaica	JAM	36	65.5 %
Philippines	PHL	36	65.5 %
Pakistan	PAK	42	76.4 %
Kenya	KEN	50	90.9 %
Nigeria	NGA	50	90.9 %



Items with the most significant tests
Timid (20%)
Shy (18.2%)
Hardworking (18.2%)
Silent (16.4%)

Items with the fewest tests
Good-natured (1.8%)
Respectful (1.8%)
Honest (3.6%)
Cheerful (5.5%)

Results within each nation: Canada

Nation	ISO3	number_significant	proportion
Australia	AUS	2	3.6 %
United Kingdom	GBR	12	21.8 %
USA	USA	15	27.3 %
South Africa	ZAF	29	52.7 %
Trinidad and Tobago	TTO	33	60 %
Namibia	NAM	35	63.6 %
Jamaica	JAM	42	76.4 %
Philippines	PHL	43	78.2 %
India	IND	45	81.8 %
Pakistan	PAK	48	87.3 %
Nigeria	NGA	49	89.1 %
Kenya	KEN	52	94.5 %



Items with the most significant tests
Weak (18.2%)
Silent (18.2%)
Shy (18.2%)
Quiet (18.2%)

Items with the fewest tests
Active (3.6%)
Compassionate (7.3%)
Honest (7.3%)
Humane (4.3%)

Results within each nation: India

Nation	ISO3	number_significant	proportion
Trinidad and Tobago	TTO	34	61.8 %
Australia	AUS	36	65.5 %
Namibia	NAM	36	65.5 %
South Africa	ZAF	36	65.5 %
Pakistan	PAK	38	69.1 %
Jamaica	JAM	41	74.5 %
Canada	CAN	45	81.8 %
Nigeria	NGA	48	87.3 %
United Kingdom	GBR	52	94.5 %
USA	USA	55	100 %
Kenya	KEN	55	100 %
Philippines	PHL	55	100 %

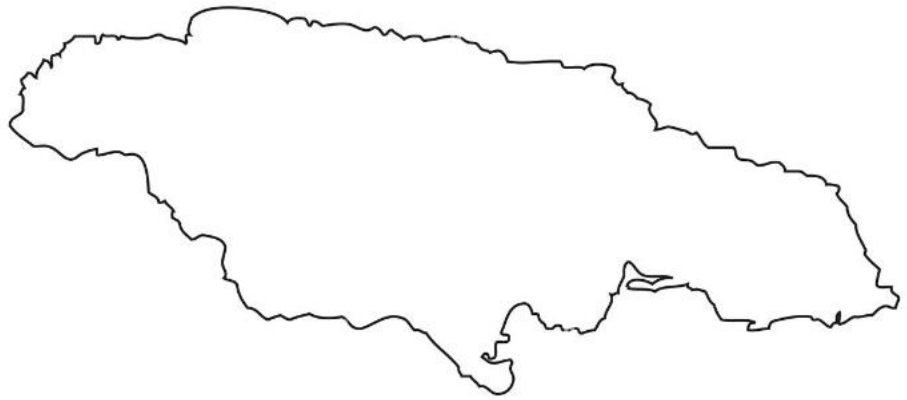


Items with the most significant tests
Weak (21.8%)
Peaceful (21.8%)
Naive (21.8%)
Hot-tempered (21.8%)

Items with the fewest tests
Compassionate (7.3%)
Humane (9.1%)
Active (10.9%)
Dynamic(10.9%)

Results within each nation: Jamaica

Nation	ISO3	number_significant	proportion
Trinidad and Tobago	TTO	18	32.7 %
Namibia	NAM	26	47.3 %
South Africa	ZAF	30	54.5 %
Australia	AUS	36	65.5 %
Nigeria	NGA	37	67.3 %
India	IND	41	74.5 %
Canada	CAN	42	76.4 %
Kenya	KEN	43	78.2 %
Pakistan	PAK	43	78.2 %
United Kingdom	GBR	48	87.3 %
USA	USA	52	94.5 %
Philippines	PHL	53	96.4 %



Items with the most significant tests
Peaceful (21.8%)
Gossipy (21.8%)
Friendly (21.8%)
Courageous (21.8%)

Items with the fewest tests
Compassionate (7.3%)
Courteous (7.3%)
Honest (7.3%)
Diligent (9.1%)

Results within each nation: Kenya

Nation	ISO3	number_significant	proportion
Nigeria	NGA	30	54.5 %
Namibia	NAM	32	58.2 %
Jamaica	JAM	43	78.2 %
Pakistan	PAK	44	80 %
South Africa	ZAF	47	85.5 %
Trinidad and Tobago	TTO	47	85.5 %
Australia	AUS	50	90.9 %
Canada	CAN	52	94.5 %
USA	USA	55	100 %
India	IND	55	100 %
Philippines	PHL	55	100 %
United Kingdom	GBR	55	100 %

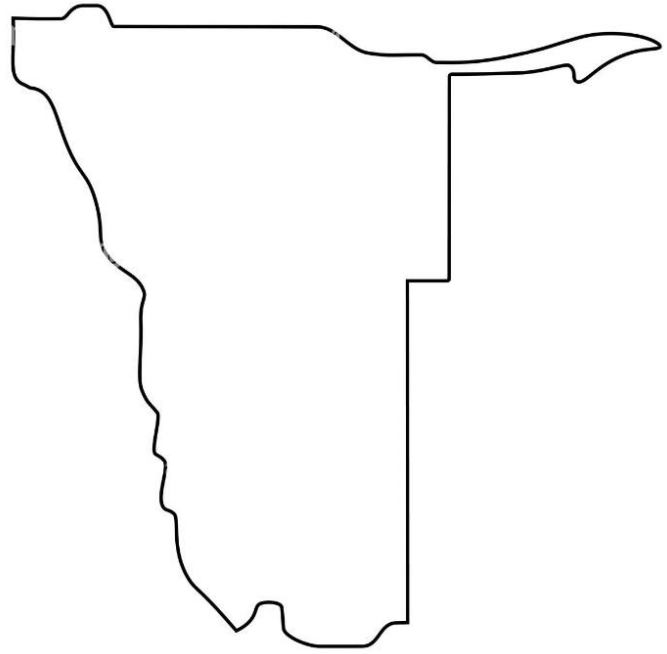


Items with the most significant tests
Silent (21.8%)
Sad (21.8%)
Naïve (21.8%)
Industrious (21.8%)

Items with the fewest tests
Faithful (10.9%)
Hardworking (10.9%)
Kind (12.7%)
Respectful (12.7%)

Results within each nation: Namibia

Nation	ISO3	number_significant	proportion
South Africa	ZAF	11	20 %
Trinidad and Tobago	TTO	21	38.2 %
Nigeria	NGA	23	41.8 %
Pakistan	PAK	24	43.6 %
Jamaica	JAM	26	47.3 %
Australia	AUS	28	50.9 %
Kenya	KEN	32	58.2 %
Canada	CAN	35	63.6 %
India	IND	36	65.5 %
United Kingdom	GBR	40	72.7 %
USA	USA	42	76.4 %
Philippines	PHL	49	89.1 %

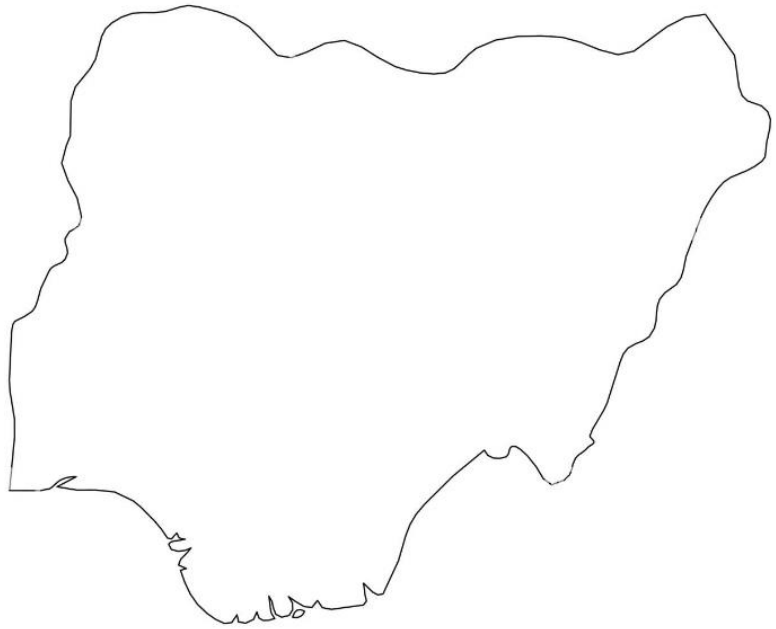


Items with the most significant tests
Friendly (20%)
Courteous (20%)
Modest (18.2%)
Disobedient (18.2%)

Items with the fewest tests
Cheerful (3.6%)
Active (5.5%)
Compassionate (5.5%)
Hardworking (5.5%)

Results within each nation: Nigeria

Nation	ISO3	number_significant	proportion
Namibia	NAM	23	41.8 %
Kenya	KEN	30	54.5 %
South Africa	ZAF	35	63.6 %
Pakistan	PAK	36	65.5 %
Jamaica	JAM	37	67.3 %
Trinidad and Tobago	TTO	38	69.1 %
India	IND	48	87.3 %
Canada	CAN	49	89.1 %
Australia	AUS	50	90.9 %
United Kingdom	GBR	52	94.5 %
USA	USA	54	98.2 %
Philippines	PHL	55	100 %



Items with the most significant tests
Modest (21.8%)
Dynamic (21.8%)
Clever (21.8%)
Clam (21.8%)

Items with the fewest tests
Hardworking (7.3%)
Honest (9.1%)
Humane (9.1%)
Kind (9.1%)

Results within each nation: Pakistan

Nation	ISO3	number_significant	proportion
Namibia	NAM	24	43.6 %
Trinidad and Tobago	TTO	33	60 %
Nigeria	NGA	36	65.5 %
India	IND	38	69.1 %
South Africa	ZAF	38	69.1 %
Australia	AUS	42	76.4 %
Jamaica	JAM	43	78.2 %
Kenya	KEN	44	80 %
Canada	CAN	48	87.3 %
United Kingdom	GBR	51	92.7 %
USA	USA	55	100 %
Philippines	PHL	55	100 %



Items with the most significant tests
Faithful (21.8%)
Weak (21.8%)
Courageous (21.8%)
Clever (21.8%)

Items with the fewest tests
Compassionate (7.3%)
Humane (7.3%)
Courteous (9.1%)
Good-natured (9.1%)

Results within each nation: The Philippines

Nation	ISO3	number_significant	proportion
Australia	AUS	36	65.5 %
Canada	CAN	43	78.2 %
Namibia	NAM	49	89.1 %
South Africa	ZAF	49	89.1 %
United Kingdom	GBR	49	89.1 %
Trinidad and Tobago	TTO	50	90.9 %
Jamaica	JAM	53	96.4 %
USA	USA	54	98.2 %
India	IND	55	100 %
Kenya	KEN	55	100 %
Nigeria	NGA	55	100 %
Pakistan	PAK	55	100 %

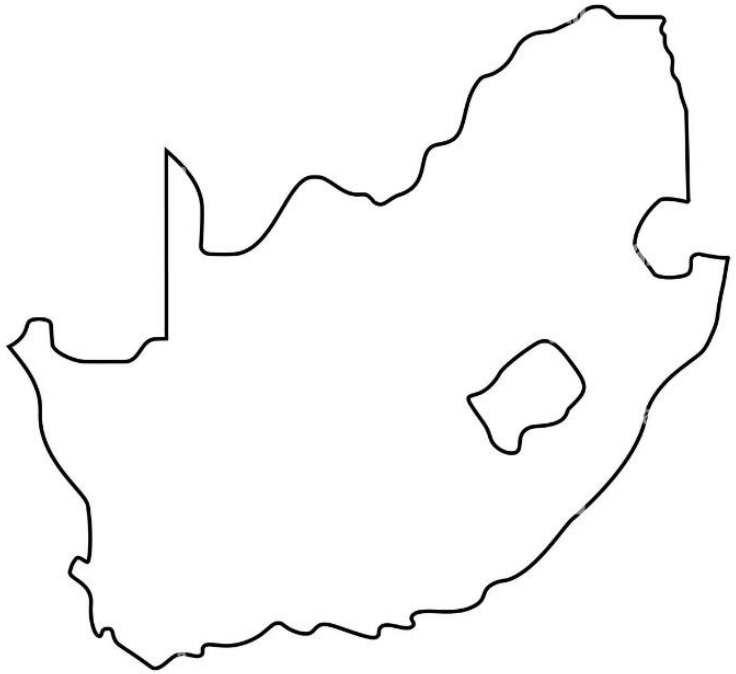


Items with the most significant tests
Weak (21.8%)
Unhappy (21.8%)
Timid (21.8%)
Silent (21.8%)

Items with the fewest tests
Courageous (14.5%)
Daring (14.5%)
Bold (16.4%)
Dynamic (16.4%)

Results within each nation: South Africa

Nation	ISO3	number_significant	proportion
Namibia	NAM	11	20 %
Australia	AUS	25	45.5 %
Trinidad and Tobago	TTO	26	47.3 %
Canada	CAN	29	52.7 %
Jamaica	JAM	30	54.5 %
Nigeria	NGA	35	63.6 %
India	IND	36	65.5 %
Pakistan	PAK	38	69.1 %
United Kingdom	GBR	38	69.1 %
USA	USA	44	80 %
Kenya	KEN	47	85.5 %
Philippines	PHL	49	89.1 %



Items with the most significant tests
Bashful (21.8%)
Silent (20%)
Sad (20%)
Quiet (20%)

Items with the fewest tests
Honest (5.5%)
Cheerful (7.3%)
Compassionate (7.3%)
Energetic (7.3%)

Results within each nation: Trinidad and Tobago

Nation	ISO3	number_significant	proportion
Jamaica	JAM	18	32.7 %
Namibia	NAM	21	38.2 %
South Africa	ZAF	26	47.3 %
Australia	AUS	32	58.2 %
Canada	CAN	33	60 %
Pakistan	PAK	33	60 %
India	IND	34	61.8 %
Nigeria	NGA	38	69.1 %
United Kingdom	GBR	44	80 %
Kenya	KEN	47	85.5 %
USA	USA	49	89.1 %
Philippines	PHL	50	90.9 %



Items with the most significant tests
Bashful (21.8%)
Hot-tempered (20%)
Happy (20%)
Friendly (20%)

Items with the fewest tests
Kind (5.5%)
Diligent (7.3%)
Honest (7.3%)
Respectful (7.3%)

Results within each nation: United Kingdom

Nation	ISO3	number_significant	proportion
Australia	AUS	9	16.4 %
Canada	CAN	12	21.8 %
USA	USA	24	43.6 %
South Africa	ZAF	38	69.1 %
Namibia	NAM	40	72.7 %
Trinidad and Tobago	TTO	44	80 %
Jamaica	JAM	48	87.3 %
Philippines	PHL	49	89.1 %
Pakistan	PAK	51	92.7 %
India	IND	52	94.5 %
Nigeria	NGA	52	94.5 %
Kenya	KEN	55	100 %

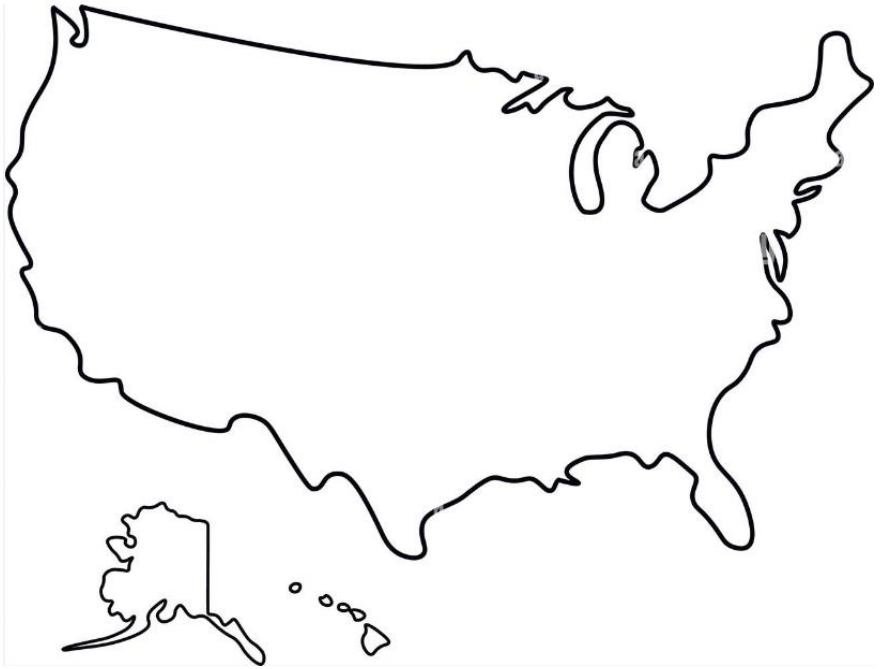


Items with the most significant tests
Timid (21.8%)
Shy (21.8%)
Aggressive (21.8%)
Silent (20%)

Items with the fewest tests
Humane(7.3%)
Clever (9.1%)
Compassionate (9.1%)
Honest (9.1%)

Results within each nation: United States

Nation	ISO3	number_significant	proportion
Australia	AUS	12	21.8 %
Canada	CAN	15	27.3 %
United Kingdom	GBR	24	43.6 %
Namibia	NAM	42	76.4 %
South Africa	ZAF	44	80 %
Trinidad and Tobago	TTO	49	89.1 %
Jamaica	JAM	52	94.5 %
Nigeria	NGA	54	98.2 %
Philippines	PHL	54	98.2 %
India	IND	55	100 %
Kenya	KEN	55	100 %
Pakistan	PAK	55	100 %



Items with the most significant tests
Weak (21.8%)
Peaceful (21.8%)
Timid (20%)
Gentle (20%)

Items with the fewest tests
Clever (10.9%)
Benevolent (12.7%)
Good-natured (12.7%)
Respectful (12.7%)